

MFFT: an improved method for stable ID tracking and counting of multi-class mango segmentation with multi-feature fusion in complex occlusion scenarios

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ABSTRACT

Accurate mango segmentation, counting, and biomass estimation are critical for orchard yield prediction. To address the challenges of low fruit segmentation accuracy, frequent ID switches, and large tracking and counting errors under complex occlusion scenarios, this paper proposes a multi-feature fusion tracking method termed MFFT. The method is built on the Swin-T backbone and a customized Transformer decoder, integrated with shallow feature enhancement, boundary attention, and background gating modules. For mango segmentation, the proposed approach achieves 0.71 mIoU and 40.32 FPS for both on-tree and fallen mangoes. In the tracking stage, IOU, HSV, shape, and diagonal features are fused, together with an improved Kalman filter and ID lifecycle management, resulting in a MOTA of 0.6513 and a significant reduction in ID switches. The system innovatively introduces diagonal length as a computational metric, achieving a counting relative error of only 4.30% and a biomass estimation error of 4.49%. It outperforms mainstream algorithms such as Mask R-CNN, YOLOv8, DTER, and ByteTrack across multiple metrics. The proposed method exhibits strong application potential in orchards and provides more precise data support and a decision-making foundation for digital agricultural production management. The code and dataset for this study have been made open source on Google Drive: <https://drive.google.com/file/d/1UzvgigqLXn-C2aJZsHmCLAWSYaVYA0bnH/view?usp=sharing>.

1. Introduction

Mango, a high-value cash crop widely cultivated in tropical and subtropical regions worldwide (Lei et al., 2025), benefits from yield prediction and precise biomass estimation, which are directly linked to the optimization of orchard irrigation and fertilization management, harvesting schedule planning, and market supply-demand regulation. These aspects form the core driver for advancing the mango industry towards large-scale and intelligent transformation (Xie et al., 2024). With the rapid development of digital agricultural technologies, computer vision, owing to its advantages of non-contact, real-time, and high-precision operation (Zhou, et al., 2025a,2025b), has gradually replaced

traditional manual counting and sampling estimation methods, emerging as a core technological pathway for monitoring crop growth status and evaluating yield in orchards (Karabanov et al., 2025). Currently, computer vision is applied in disease differentiation, fruit segmentation, yield prediction, and other areas, providing growers with comprehensive dynamic growth data of fruits throughout the entire growth cycle (Lv et al., 2025). This offers robust data support for agricultural management decision-making, holding significant practical relevance.

In recent years, extensive research on visual detection technologies for orchard crops has been conducted by scholars globally, with fruit segmentation serving as a critical preprocessing step. The

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methodological evolution in this domain has crystallized into two principal technical pathways: traditional machine vision and deep learning. Traditional approaches rely on manually designed color thresholds, texture features, and morphological operations to achieve object segmentation (Rojas Santelices et al., 2025). For example, such methods have been applied to citrus shape detection, offering simplicity and low cost. However, their applicability is confined to specific experimental conditions, lacking the robustness required for complex real-world orchard environments, which has led to their gradual displacement by deep learning techniques in numerous application scenarios (Yang et al., 2025). Among deep learning based segmentation methods, instance segmentation (Xu et al., 2025) and semantic segmentation (Xu et al., 2025) currently represent the mainstream frameworks. Nevertheless, both exhibit significant adaptation limitations when confronted with the complexities of mango orchard environments and scenarios characterized by limited sample sizes, thereby failing to meet the demands of practical deployment.

Instance segmentation stands as a primary technique for precise extraction of individual fruits, widely applied in orchard crop detection. Traditional instance segmentation models, represented by Mask R-CNN, enable end-to-end training for detection and mask generation. To address fruit segmentation under dense canopy occlusion, Liao improved the mask head of Mask R-CNN, achieving a peak accuracy of 0.941; under heavier occlusion, the model attained a best recall of 0.846 (Liao et al., 2025). Crespo employed Mask R-CNN with a ResNet-FPN backbone from Detectron2 for strawberry instance segmentation, obtaining an mAP of 0.8345 (Crespo et al., 2025). Moreover, Zu and Lee respectively integrated ResNet50 (Zu et al., 2021) and ResNet101 (Lee et al., 2020) with Mask R-CNN, both reaching high accuracy on their respective datasets. However, these models are complex and generally exhibit slow inference speeds. In subsequent developments, YOLO has proven to be a highly efficient detection-segmentation architecture, widely adopted for fruit segmentation tasks. Zhou applied an improved YOLO v8 for instance segmentation and localization of multi-colored apples, achieving mAP above 0.95 for each color variant (Zhou, et al., 2025a,2025b). For grape-peduncle segmentation, Shen designed MSA-YOLO based on YOLOv8, attaining a segmentation mAP@50 of 0.995 (Shen et al., 2025). To adapt to complex environments, Wei refined YOLOv8, improving segmentation mask mAP50 and mAP50-95 by 5.8% and 8.5% respectively over the original version (Wei et al., 2025). Among recent advanced models, DETR, with its Transformer-based global feature capture capability, provides more precise boundary delineation and is gaining increasing attention. Zhu utilized a custom transformer to enhance segmentation AP (Zhu et al., 2025a). Zhu employed an improved mask2former for watermelon-scion leaf segmentation, achieving an mIOU of 0.9744 (Zhu et al., 2025b). In contrast to instance segmentation, semantic segmentation networks adopt lightweight encoder-decoder structures, offering efficient inference and strong hardware compatibility. They enable rapid pixel-wise separation of fruit from background, effectively handling complex background interference. Ibarra evaluated UNet for segmenting tomato fruits and leaves, reporting a peak performance of 0.89078 (Guerra Ibarra et al., 2025). Mondal used MobileNet-v2 as the backbone with UNet as the image segmentation model, obtaining IoU and Dice scores of 95.69% and 97.89% respectively (Mondal et al., 2025). Liu designed a fish segmentation model that reached an mIOU of 0.9273 (Liu et al., 2024a). However, in mango orchard scenarios characterized by complex viewing-angle and illumination variations, extensive bagging, dense clustering and adhesion of fruits, and severe occlusion among leaves and mangoes, prior research remains limited. In Swin-S and SegFormer achieved the best IoU (67.35% and 66.44%, respectively) for clustered mango-flower segmentation, yet their mAP for isolated mangoes was inferior to that of YOLO (Ven et al., 2025).

Following fruit segmentation, the subsequent step is tracking and counting, with the core methodology being multi-object tracking (MOT). Existing MOT algorithms, such as SORT and DeepSORT,

predominantly operate in an online manner, relying on segmentation or detection results and achieving identity association through feature matching. Zhang employed YOLOv8 and DeepSORT for pedestrian detection and tracking, with improvements yielding a 9.7% increase in MOTA (Zhang, et al., 2025a,2025b). Sim applied StrongSORT to multi-cow tracking, achieving HOTA, MOTA, and IDF1 scores of 78.64%, 90.29%, and 91.41%, respectively (Sim and Cho, 2025). Harefa integrated the efficient IoU-based matching of ByteTrack with the appearance-driven association of DeepSORT, resulting in significant improvements in MOTA and IDF1 compared to using either method independently (Harefa and Wijaya, 2025). Some researchers have extracted core concepts and developed customized trackers by fusing different features for identity matching, tailored to specific practical conditions. Zhou developed a MultiMap component combined with existing advanced tracking algorithms to achieve real-time tracking of multiple fruits, with the lowest counting error rate reaching only 6.7% (Zhou et al., 2024). Liu addressed occlusion and deformation in fish movement by leveraging appearance information and motion models, enabling multi-fish tracking with a HOTA score of 71.4% (Liu et al., 2024b). Yao designed an end-to-end real-time fish tracking model, FMRFT, achieving an IDF1 of 90.3% and a MOTA accuracy of 94.3% (Yao et al., 2025). However, in the context of mango orchards, few studies have investigated tracking and counting methods under such complex conditions. Pichhika proposed a centroid-tracking algorithm for precise counting (Pichhika et al., 2025), but it is unsuitable for scenarios with severe mutual occlusion among mangoes. Previous tracker designs that fuse different features for specific scenarios provide valuable insights for subsequent development.

The final stage is biomass estimation, which calculates the theoretical fruit yield in an orchard, reduces labor input, and provides direct support for orchard production. Some studies have already employed detection and counting methods for biomass estimation. For example, the passion fruit yield was estimated using the YOLOv8n + OC-SORT + CRCM algorithm, achieving an average counting accuracy of 87.0% and effectively addressing issues such as false positives, missed detections of small fruits, and duplicate counting (Tu et al., 2025). However, this approach lacks correlation with actual yield. Another study, YOLO-F-CAP, enhanced the detection capability of YOLO for citrus in complex environments but did not incorporate continuous counting (Zeng et al., 2025). These examples illustrate that current research on biomass estimation has not yet effectively integrated segmentation results with counting outcomes.

Inspired by the above, this study formulates integrated methods for segmentation, tracking-counting, and biomass estimation to address the current challenges in mango orchard segmentation, tracking, and biomass assessment. The main contributions of this work are as follows:

- (1) A segmentation method adapted to small-sample complex scenarios is constructed. To overcome the shortcomings of existing segmentation approaches, a pixel-wise camouflaged instance network is designed, integrating a Swin-T backbone with a custom Transformer decoder and multiple specialized modules. This network alleviates the problem of dense fruit adhesion, enables segmentation of both on-tree and on-ground mangoes, and improves segmentation accuracy under small-sample training with complex data.
- (2) A tracking-counting method based on multi-feature fusion and ID lifecycle management is proposed. Features including IoU, Hu moments, HSV color, and mango diagonal length are fused to enhance cross-frame association stability. This is combined with an ID lifecycle management mechanism to reduce issues of duplicate counting, target loss, and ID switches.
- (3) A complete pipeline from segmentation to tracking to biomass estimation is established. By using the mango diagonal length extracted from segmentation as a core feature and integrating it

with tracking–counting results, the correlation between biomass assessment and actual yield is strengthened.

2. Materials and methods

2.1. Data collection

The dataset for this study was collected from a 0.3-hectare commercial Jinhuang mango plantation in Baoting County, Hainan Province ($18^{\circ}23'–18^{\circ}53'N$, $109^{\circ}21'–109^{\circ}48'E$). Specific counting regions were predefined based on the orchard layout, taking single rows as basic units to delineate Regions of Interest (ROIs), each assigned a unique identifier for precise association of subsequent counting results with target trees or rows. Since the photographs included a large number of blurry mangoes, to facilitate statistical analysis, the count was limited to clearly identifiable mangoes—both those hanging on trees in the foreground of the frame and those that had fallen to the ground. Mangoes that were too far in the background or too high in the tree canopy to be clearly identified were excluded from this count, thereby enhancing the validity and reliability of the results in this study. Data acquisition was conducted during the ripening and harvesting periods (March 2025) using a smartphone (50MP main camera with 24 mm focal length, 48MP ultra-wide-angle camera with 13 mm focal length), set to 720×1280 pixel resolution, auto-adjusted ISO (100–800) and white balance, and 1/60–1/1000 s shutter speed to avoid motion blur. Multiple viewing angles were systematically employed, including ground-level view and inter-row side view, comprehensively capturing the visual characteristics of both hanging mangoes (with/without occlusion, clustered stacking) and fallen mangoes on the ground (isolated/stacked) against complex backgrounds (soil, weeds, branches, leaves). The acquisition covered various lighting conditions (sunny, cloudy, weak light, dusk) and different times of day (morning, afternoon, evening) to ensure scene diversity (Fig. 1).

Annotations were performed at the pixel level using the LabelMe tool, with two defined categories: on-tree mangoes and on-ground mangoes. By simultaneously monitoring both tree-ripened and ground-fallen mangoes, this approach enables a comprehensive assessment of growth status, fruit drop and loss, and disease risk, providing data to support precision management of orchards and yield forecasting. Due to continuous variations in camera angle, distance, and fruit pose during video recording (30 FPS), key frames were extracted at 2 FPS, and valid images were screened to exclude motion-blurred or over/underexposed ones, ensuring each image contained at least 10 mango instances. Polygons were used to outline mangoes, prioritizing clearly visible and fully shaped fruits to avoid mutual interference. A total of 552 valid images were annotated, with a random 8:2 split resulting in 439 training images and 113 validation images—both sets maintaining

consistent distributions of regions, environments, angles, and mango densities. Separate JSON-format annotation files for the two sets were generated by LabelMe, containing category, image size, bounding boxes, and masks. To meet segmentation network input requirements and optimize training efficiency, all images were uniformly resized to 256×256 pixels, and ImageNet-standard normalization (mean [0.485, 0.456, 0.406], standard deviation [0.229, 0.224, 0.225]) was applied to standardize pixel value distributions, enhancing training stability and convergence speed.

It is worth noting that bagging cultivation is widely adopted in major mango production areas. This practice improves fruit appearance quality and reduces pest and disease damage. This study focuses on bagged mangoes as the research subject. This setting is highly consistent with actual production standards. The remaining challenges include branch and leaf occlusion, fruit stacking, and uneven illumination. These factors impose realistic demands on the robustness of orchard vision algorithms. The research outcomes can directly support automated management in mainstream production scenarios. All samples in this study were collected from naturally bagged mangoes that met standard orchard practices. The bagging operations followed local cultivation regulations. No bags were removed prior to data collection. This ensures that the dataset maintains consistency with real operational environments.

2.2. Segmented network architecture design

The proposed mango segmentation model is constructed on a Swin Transformer-Tiny (Swin-T) backbone and a custom Transformer decoder. To address challenges such as small-size mango targets, dense clustering and adhesion, and complex backgrounds, multiple dedicated modules are introduced. The overall architecture is illustrated in Fig. 2. The input mango image is processed by the Swin-T backbone to extract hierarchical features and semantic information. Subsequently, the features are fed into a Shallow-Feature Enhancement (SFE) module to achieve effective fusion of shallow detail and deep semantic features. Within the decoder, the features first pass through a feature projection layer and a coordinate attention block, followed by multi-scale feature fusion. The fused features then sequentially flow through three enhancement modules: Spatial-Orientation Attention (SOA), Boundary-Aware Module (BAM), and Background-Global Modulation (BGM). Mask aggregation is subsequently performed. Finally, the features are processed by a Mask Head for dimension adjustment and upsampled to match the input resolution, followed by a mask-refinement module to output the refined mango segmentation mask. The details of these modules are described in the following sections.

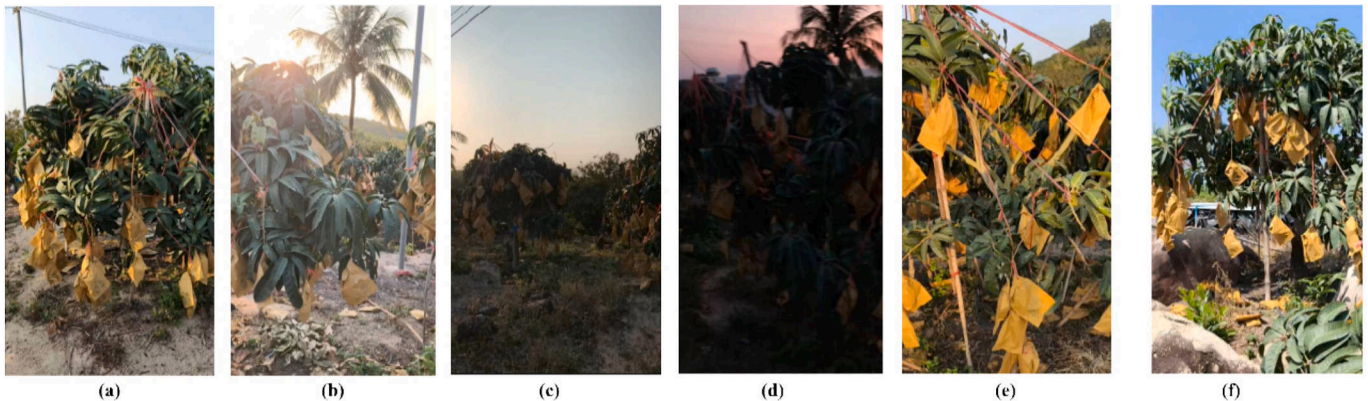


Fig. 1. The mango data collection environment includes: (a) Normal lighting (b) Strong light (c) Weak light (d) Dusk (extremely weak light) (e) Dense shading (f) Complex backgrounds.

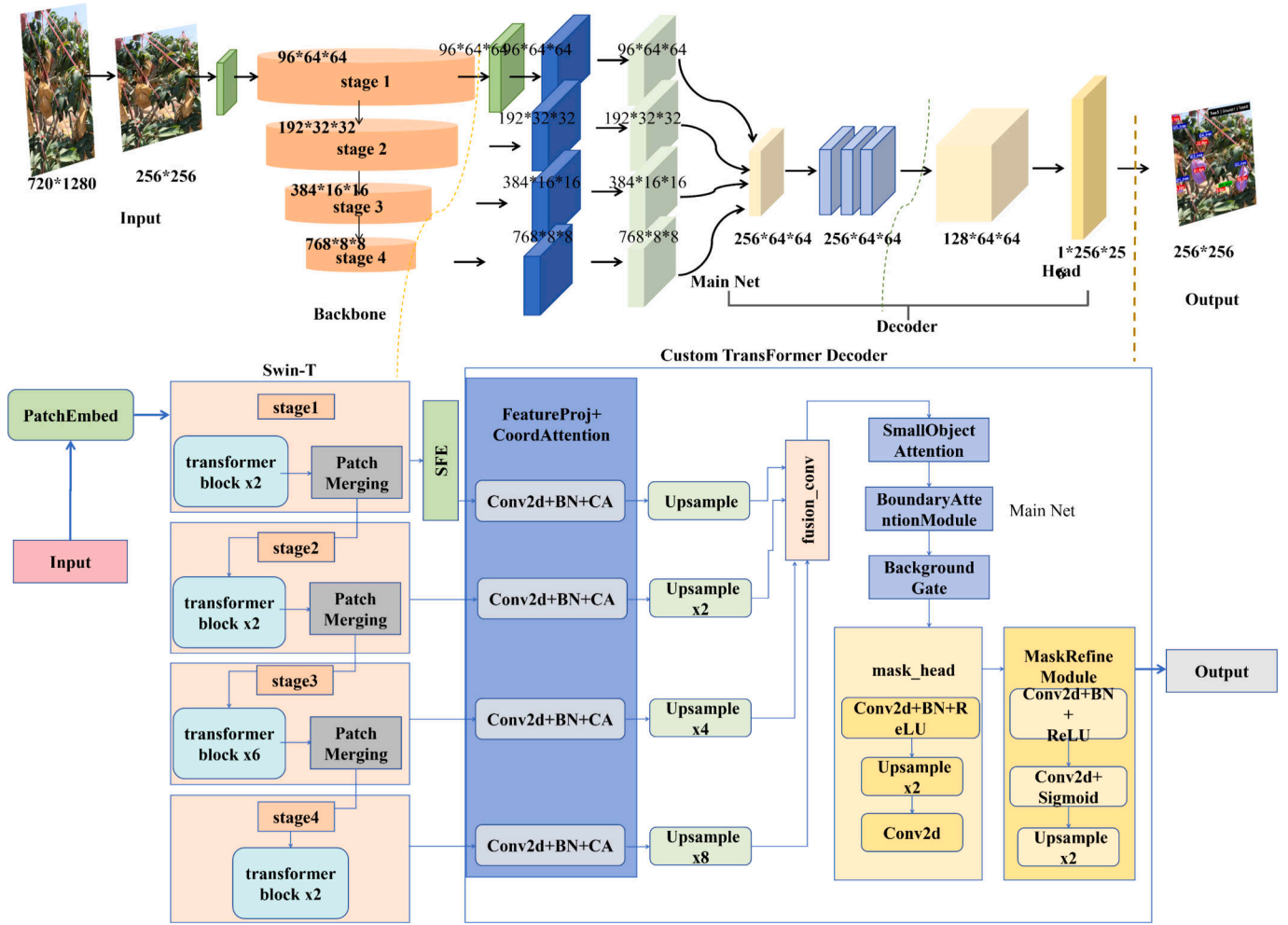


Fig. 2. Overall Architecture of the Mango Segmentation Network.

2.2.1. Stem feature extraction network (Swin-T)

Swin Transformer (Swin-T) is built upon the Transformer-based vision paradigm, employing a local window self-attention mechanism to efficiently process multi-scale features. Compared to traditional CNNs, its self-attention mechanism effectively captures long-range dependencies and exhibits strong generalization capabilities, making it suitable for large-scale image tasks. In this study, a pre-trained Swin-T model is adopted as the backbone network to achieve multi-scale feature representation of mango targets. The feature extraction process is as follows: a 256×256 input image is divided into 4×4 patches via the Patch Embedding module and linearly projected into a feature map of $96 \times 64 \times 64$. It then passes through four stages of Swin Transformer Blocks and Patch Merging operations, progressively downsampling while increasing the channel dimensions: Stage 1 outputs $96 \times 64 \times 64$, Stage 2 outputs $192 \times 32 \times 32$, Stage 3 outputs $384 \times 16 \times 16$, and Stage 4 outputs $768 \times 8 \times 8$. To adapt to transfer learning and prevent overfitting, a staged unfreezing strategy is applied: initially, Stages 1–2 are frozen while only Stages 3–4 and the decoder are trained; Stage 2 is unfrozen at epoch 15, and Stage 1 is unfrozen at epoch 25. This approach balances the transfer of pre-trained features with scene adaptation, thereby enhancing training stability and feature representation accuracy.

2.2.2. Custom transformer decoder

Mask2Former, a high-performance segmentation model, integrates the core advantages of Mask Transformer and convolutional neural networks to achieve precise object segmentation. Its key innovation lies

in transforming pixel-wise segmentation into a sequence modeling task, leveraging the Transformer architecture for feature aggregation and mask prediction. This design combines multi-scale information processing capability with the robustness and accuracy improvements provided by the self-attention mechanism. In the present model, the framework and structure of Mask2Former's decoder are adapted to design a custom Transformer decoder. The decoder is deeply coupled with the backbone, preserving benefits such as multi-scale feature fusion and self-attention, while integrating multiple attention and enhancement modules to improve the accuracy of mango mask predictions, thereby enhancing segmentation performance in complex mango orchard environments.

Shallow feature enhancement

To mitigate the inherent trade-off in deep convolutional networks—namely, insufficient semantic information in shallow features and loss of spatial detail in deep features—this paper designs the Shallow Feature Enhancement (SFE) module, as illustrated in Fig. 3. The module adopts a cross-scale fusion strategy: deep features are upsampled and aligned with shallow features, and both are then projected and concatenated along the channel dimension. A subsequent 3×3 convolution performs local association modeling, yielding enhanced features that preserve both fine detail and rich semantics. This design effectively improves feature representation for small targets.

Feature projection layer and coordinate attention

To unify multi-scale feature dimensions and enhance spatial locali-

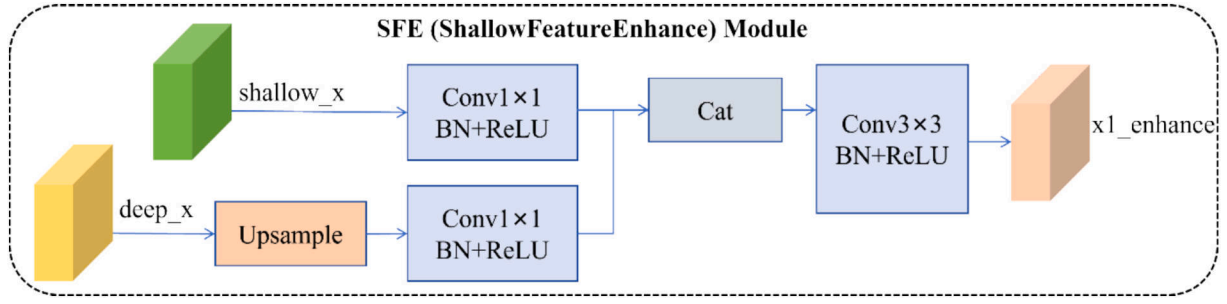


Fig. 3. Structure of the SFE Module.

zation capability, this study integrates a feature projection layer with a coordinate attention mechanism. The module first reduces the dimensionality of multi-scale inputs via 1×1 convolutions, then incorporates coordinate attention to capture spatial distribution information of the target, thereby compensating for the limitation of traditional projection that focuses solely on channel-wise features. The projected features undergo bidirectional coordinate encoding, feature concatenation, and weight generation, achieving joint spatial-channel modulation and thereby improving feature representation capability (Eqs. 1-6).

$$X_{proj} = \text{BN}(\text{Conv}_{1 \times 1}(X)) \quad (1)$$

$$\begin{aligned} X_h &= \text{AdaptiveAvgPool2d}(X_{proj}, (H, 1)), X_w \\ &= \text{AdaptiveAvgPool2d}(X_{proj}, (1, W)) \end{aligned} \quad (2)$$

$$X_{cat} = \text{ReLU}(\text{BN}(\text{Conv}_{1 \times 1}([X_h; X_w^T]))) \quad (3)$$

$$X_{h'}, X_{w'} = \text{Split}(X_{cat}) \quad (4)$$

$$A_h = \text{Sigmoid}(\text{Conv}_{1 \times 1}(X_{h'})), \quad A_w = \text{Sigmoid}(\text{Conv}_{1 \times 1}(X_{w'})) \quad (5)$$

$$X_{out} = X_{proj} \odot A_h \odot A_w \quad (6)$$

where, X denotes the input feature map, X_{proj} represents the feature after the projection layer, X_h corresponds to the vertical-direction information of the projected features, X_w indicates the horizontal-direction information of the projected features, X_{cat} refers to the concatenated and transformed features, $X_{h'}$ stands for the split vertical-direction information, $X_{w'}$ signifies the split horizontal-direction information, A_h is the vertical attention weight, A_w is the horizontal attention weight, and X_{out} denotes the output feature map.

Multi-scale feature fusion

The multi-scale feature fusion module aligns feature maps from different scales (Stage 2-4) to a unified spatial resolution via bilinear up-sampling, then concatenates them with the Stage 1 feature map along the channel dimension. A subsequent 3×3 convolution reduces the concatenated features to 256 channels, preserving semantic information and fine detail from each scale. This design enables the model to extract

rich, multi-scale representations, enhancing its ability to recognize mango targets of varying sizes and levels of detail, and proving particularly effective in handling both large and small objects.

SOA small objective attention module (SOA)

To enhance the detection performance for small-sized mango targets (such as young or occluded fruits), a small-target attention module (Fig. 4) is designed. The module adopts a dual-branch architecture: the spatial detail branch employs 1×1 convolutions to extract local features and integrates dilated convolutions (dilation rate = 2) to expand the receptive field for contextual information incorporation; the channel enhancement branch generates channel weights via adaptive pooling and scaling operations to reinforce small-target relevant features and suppress background interference. The two-stream features are normalized via Batch Normalization (BN) and then fused, effectively mitigating the issue of small-target feature suppression (Eqs. 7-9).

$$X_{fused} = \text{BN}(\text{Conv}_{1 \times 1}(X) + \text{Conv}_{3 \times 3(d=2)}(X)) \quad (7)$$

$$A = \text{Sigmoid}(\text{Conv}_{1 \times 1}(\text{ReLU}(\text{Conv}_{1 \times 1}(\text{AdaptiveAvgPool2d}(X_{fused})))) \quad (8)$$

$$X_{out} = X_{fused} \odot A + X \quad (9)$$

where X_{fused} represents the fused feature map, A corresponds to the attention weight map, and X_{output} indicates the output enhanced feature map.

BAM boundary attention module (BAM)

To address the challenge of blurred boundaries in overlapping mango regions that complicates segmentation, this paper designs a Boundary Attention Module (Fig. 5), which enhances boundary perception through multi-scale edge feature fusion. The module employs three parallel convolutional layers with dilation rates of 1, 2, and 4 to extract fine, medium, and coarse-grained edge features, respectively. After batch normalization and ReLU activation, the features are concatenated. A 1×1 convolution followed by a sigmoid function then generates an edge mask to highlight boundary regions. Simultaneously, a channel attention mechanism is incorporated to adaptively enhance

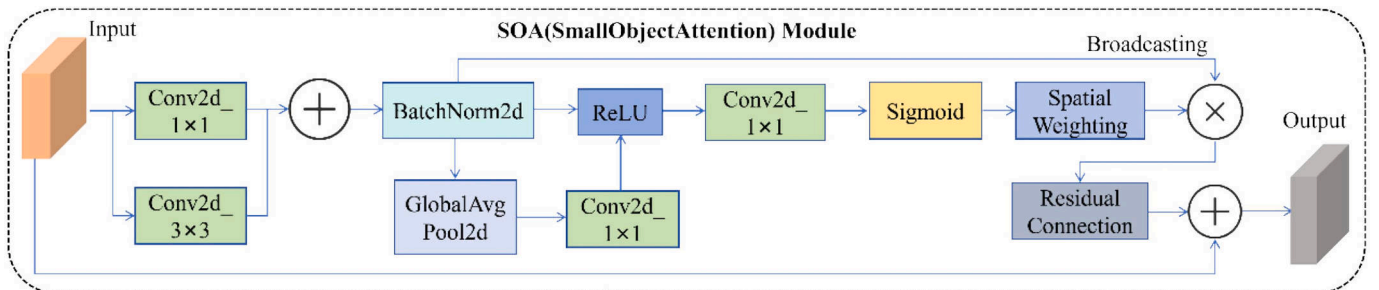


Fig. 4. Structure of the Small Target Attention Module.

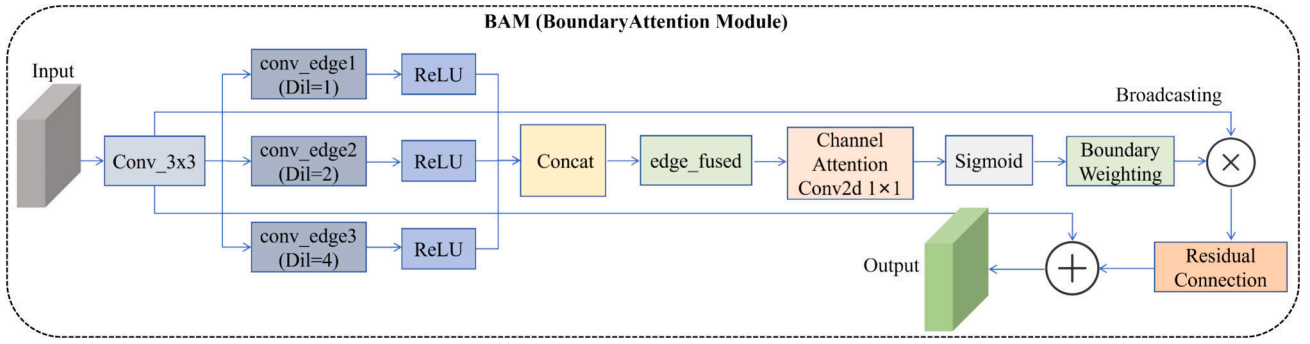


Fig. 5. Structure of the Boundary Attention Module.

the response weights of boundary-relevant feature channels. This design effectively captures fine boundary details in adherent regions, significantly improving segmentation accuracy (Eqs. 10-14).

$$\begin{aligned} E_1 &= \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3(d=1)}(X))) \\ E_2 &= \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3(d=2)}(X))) \\ E_3 &= \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3(d=4)}(X))) \end{aligned} \quad (10)$$

$$M_{\text{edge}} = \text{Sigmoid}(\text{Conv}_{3 \times 3}([E_1; E_2; E_3])) \quad (11)$$

$$X_{\text{edge}} = X \odot M_{\text{edge}}, \quad A = \text{ChannelAttention}(X_{\text{edge}}) \quad (12)$$

$$X_{\text{att}} = X_{\text{edge}} \odot A \quad (13)$$

$$X_{\text{out}} = X_{\text{att}} + X \quad (14)$$

Where, E_1, E_2, E_3 represents the edge feature maps obtained via three different dilation rates; M_{edge} corresponds to the edge attention weights; X_{edge} indicates the edge-enhanced feature map; A signifies the channel-wise attention weights; X_{att} refers to the feature map incorporating both spatial edge enhancement and channel importance weighting; and X_{out} denotes the output feature map.

Background suppression module

The background gating module employs a 1×1 convolution, batch normalization (BN), ReLU activation, and a sigmoid function to generate a gating weight (Fig. 6). This weight is multiplied with the feature map, effectively suppressing background features while enhancing the saliency of foreground targets (mangoes). By increasing the contrast between targets and background and reducing the impact of background noise, the module enables the network to focus more precisely on mango regions, thereby improving segmentation accuracy.

Masked aggregation module

The module aims to address the feature confusion caused by densely clustered and adhering mangoes by using segmentation masks to guide feature weighting and aggregation, thereby enhancing the feature independence of individual mango masks. A 1×1 convolution ($\text{Conv}_{1 \times 1}$)

$+ \text{BN} + \text{ReLU}$) is applied to reduce dimensionality and normalize the input features. Adaptive average pooling and max pooling are then used to extract instance-core features, followed by another 1×1 convolution to fuse the dual-pooling outputs. Simultaneously, a mask dimension-extension and interpolation strategy is designed to align the mask size with the feature map dimensions (Eqs. 15-20).

$$X_{\text{proj}} = \text{ReLU}(\text{BN}(\text{Conv}_{1 \times 1}(X))) \quad (15)$$

$$M' = \text{Interpolate}(M, \text{size} = X_{\text{proj}}.\text{shape}[2 :]) \quad (16)$$

$$X_{\text{masked}} = X_{\text{proj}} \odot M' \quad (17)$$

$$\begin{aligned} X_{\text{avg}} &= \text{AdaptiveAvgPool2d}(X_{\text{masked}}), \quad X_{\text{max}} \\ &= \text{AdaptiveMaxPool2d}(X_{\text{masked}}) \end{aligned} \quad (18)$$

$$X_{\text{fused}} = \text{Conv}_{1 \times 1}([X_{\text{avg}}; X_{\text{max}}]) \quad (19)$$

$$X_{\text{agg}} = X_{\text{proj}} + \text{Expand}(X_{\text{fused}}, \text{size} = X_{\text{proj}}.\text{shape}) \quad (20)$$

where M corresponds to the externally input mask, M' indicates the spatially aligned mask, X_{masked} refers to the mask-weighted feature map, X_{avg} signifies the result of global average pooling, X_{max} denotes the result of global max pooling, X_{fused} represents the fused global feature map, and X_{agg} indicates the final aggregated feature map.

Mask prediction header and refinement module

The mask prediction head outputs the final mango segmentation mask by integrating the core semantics and boundary information of the target, extracting predictions based on pixel connectivity. Acting as a post-processing unit, the mask refinement module employs a feature enhancement, weight modulation, and residual connection architecture. Two 3×3 convolutional layers are used to enhance feature representation, supported by batch normalization and ReLU activation for training stability, with a sigmoid function generating per-pixel refinement weights at the output. A 1×1 convolutional residual branch is incorporated to preserve original features during refinement,

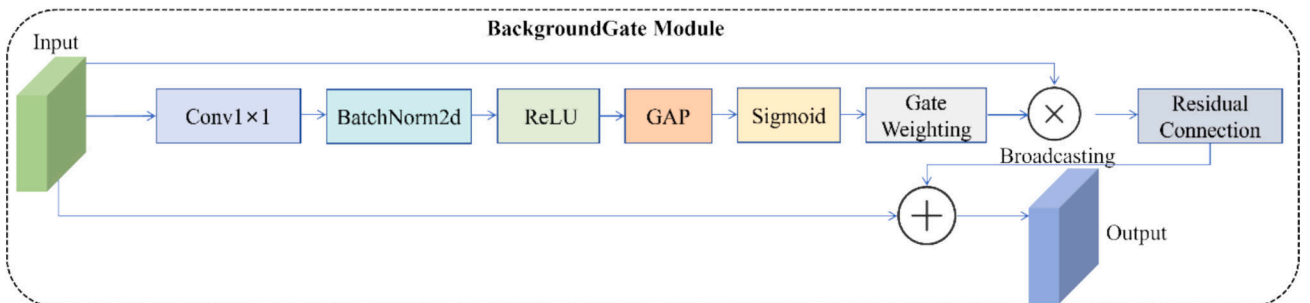


Fig. 6. Structure of the Background Suppression Module.

significantly improving mask edge sharpness and detail retention (Eqs. 21–22).

$$W = \text{Sigmoid}(\text{BN}(\text{Conv}_{3 \times 3}(\text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(X)))))) \quad (21)$$

$$(\text{Conv}_{3 \times 3}(X)))X_{\text{out}} = X \odot W + \text{Conv}_{1 \times 1}(X) \quad (22)$$

Where, X represents the input feature map, W denotes the spatial attention weights, and X_{out} signifies the output feature map.

Diagonal reasoning

The calculation of the mango diagonal length is based on the refined instance segmentation mask. First, all non-zero pixel coordinates are extracted from the binarized mask to form a point set. Principal component analysis (PCA) is applied to this set to determine the first principal component direction, which corresponds to the major axis of the mango. All pixels are then projected onto this axis, and the two pixels with the maximum and minimum projection values are selected as the endpoints of the diagonal. The diagonal length is finally computed as the Euclidean distance between these endpoints. The workflow for extracting this morphological feature is illustrated in Fig. 7 (Eqs. 23–28).

$$P = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_N & y_N \end{bmatrix}, \quad i = 1, 2, \dots, N \quad (23)$$

$$\vec{v}_1 = \text{PCA}_1(P) = [v_{1x}, v_{1y}]^T \quad (24)$$

$$\text{proj}_i = \vec{p}_i \cdot \vec{v}_1 = x_i \cdot v_{1x} + y_i \cdot v_{1y} \quad (25)$$

$$\text{id}_{\min} = \text{argmin}_i(\text{proj}_i), \quad \text{id}_{\max} = \text{argmax}_i(\text{proj}_i) \quad (26)$$

$$p_1 = (x_{\text{id}_{\min}}, y_{\text{id}_{\min}}), \quad p_2 = (x_{\text{id}_{\max}}, y_{\text{id}_{\max}}) \quad (27)$$

$$L = \sqrt{(x_{\text{id}_{\max}} - x_{\text{id}_{\min}})^2 + (y_{\text{id}_{\max}} - y_{\text{id}_{\min}})^2} \quad (28)$$

Among these, P represents the set of non-zero valid pixel coordinates for a single mango mask. $\vec{v}_1$ denotes the principal direction of the point set distribution, with v_{1x} and v_{1y} being its components along the x and y axes, respectively. $\vec{p}_i = (x_i, y_i)$ is the coordinate vector of point i within the set, and proj_i is the projection of this coordinate vector onto the principal direction. $\text{id}_{\min}$ represents the i -value of the point with the smallest projection value, while $\text{id}_{\max}$ denotes the point with the largest projection value. p_1, p_2 signifies the maximum length of the diagonal endpoint L along the principal direction, the diagonal length.

It is worth noting that during the research process, we also attempted to annotate the diagonal length of each mango simultaneously in Labelme, aiming to train a model for inferring this parameter. However, due to the difficulty in establishing a unified annotation rule for the diagonal and the lack of precise correspondence between mango instances and their diagonals in the annotations, the model learning performance was significantly poor. Consequently, this approach was ultimately abandoned.

2.2.3. Dynamic weighted hybrid loss function

To address key challenges in mango segmentation, including class imbalance, instance adhesion, and boundary blur, this study designs a dynamic segmentation loss function comprising six loss components. A focal cross-entropy loss with online hard example mining (weight 0.3) mitigates pixel-level class imbalance. A focal Dice loss (weight 0.3) enhances sensitivity to small targets and fine edges. The Lovász-Softmax loss (weight 0.2) directly optimizes the IoU metric. An instance-repulsion loss (weight 0.1) reduces merging of adjacent instances. A class-contrast loss (weight 0.05) strengthens inter-class feature discrimination, and a polygon-boundary loss (weight 0.05) refines mask edge accuracy. The joint optimization of these losses significantly improves segmentation performance in complex scenarios.

2.3. Tracking network design

The segmentation network outputs mango masks and diagonal features that provide fundamental target information for subsequent tracking and counting. However, in orchard environments, challenges such as dense fruit adhesion, dynamic occlusion by branches and leaves often lead to blurred mask boundaries and temporary target disappearance. Directly inputting these segmentation results into conventional tracking algorithms can easily cause ID switches, duplicate counts, or target loss. To address these issues, this study designs a tracker based on multi-feature fusion and ID lifecycle management. It integrates geometric, shape, and appearance features from the segmentation output, and establishes a feature matching priority mechanism. This approach enables efficient bridging between segmentation results and the tracking task, effectively handling the tracking challenges in complex orchard scenarios. The overall architecture of the tracking network is illustrated in Fig. 8.

2.3.1. Instance extraction and multi-feature computation

Based on the class mask output from semantic segmentation, mango instances are separated through morphological processing and contour analysis. The specific steps are as follows: 1) Perform erosion–dilation morphological operations on the class mask to remove salt–and–pepper noise; 2) Apply a contour detection algorithm to extract object contours, filtering out small regions with an area below 30 pixels to eliminate background noise interference; 3) Calculate the minimum bounding rectangle of each contour to obtain the object bounding box, and refine contour concavities using convex hull optimization to improve the alignment between the bounding box and the actual shape of the mango. To enhance the reliability of object association and avoid the limitations of single–feature matching, four complementary core features are extracted, as detailed below:

- (1) **Position features:** Include bounding–box coordinates (x_1, y_1, x_2, y_2) and center coordinates (c_x, c_y), which directly represent the spatial location of the target in the image and provide positional cues for initial matching.
- (2) **Shape features:** Based on Hu moments, a 6–dimensional shape descriptor is generated through logarithmic transformation and

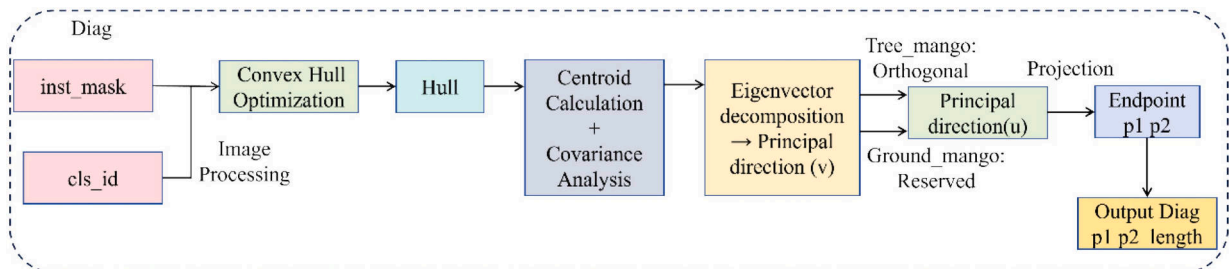


Fig. 7. Diagonal Morphological Feature Calculation Flowchart.

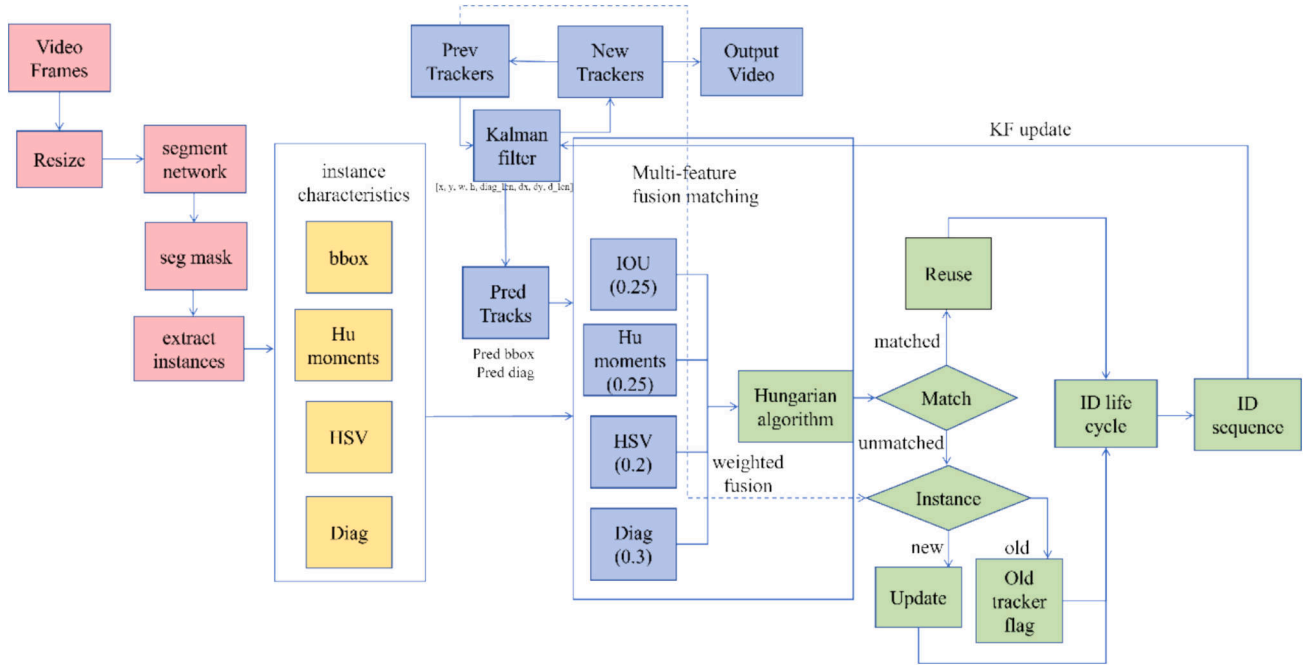


Fig. 8. Multi-Feature Fusion and ID Lifecycle Management Tracker Architecture.

sign normalization. Hu moments possess rotation, scale, and translation invariance, effectively characterizing the intrinsic shape of mangoes and reducing the impact of pose variations on matching. The shape feature vector is denoted as $[\eta_1, \eta_2, \dots, \eta_6]$, where η_i is the normalized i_{th} order Hu moment.

- (3) **Color features:** Extract the histogram of the target mask region in HSV color space, using an $18 \times 8 \times 8$ binning strategy to produce a 288-dimensional color descriptor, which is normalized to serve as the color feature. The HSV color space offers greater robustness to illumination changes compared to RGB space, effectively mitigating the influence of natural-scene lighting fluctuations on feature representation.
- (4) **Diagonal morphological feature:** Tailored to the morphological characteristics of mangoes, a diagonal calculation method based on principal-direction analysis of the convex hull contour is designed, as illustrated in Fig. 7. The principal direction of the convex hull contour is computed via covariance analysis. This direction is then adapted according to the mango category (vertical direction is prioritized for on-tree mangoes, horizontal direction for on-ground mangoes) to ensure the diagonal accurately represents the target shape. The contour is projected along the adapted principal direction to locate the extreme points (p_1, p_2) as the diagonal endpoints, and the diagonal length L is computed. This feature exhibits strong representational capacity for variations in mango size and shape, serving as a core morphological feature for improving matching reliability.

The tracker, which integrates multi-feature fusion and ID lifecycle management, merges the geometric, shape, and appearance features output by the segmentation module, establishes a feature-matching priority mechanism, and achieves efficient coupling between segmentation results and the tracking task to address the tracking challenges in complex orchard environments.

2.3.2. Improved kalman filter prediction module

Traditional Kalman filtering constructs the state vector solely based on the bounding box (x, y, w, h) and does not account for morphological stability of mangoes, making it susceptible to prediction errors in scenarios involving occlusion or subtle shape changes. This paper proposes

an Enhanced Kalman Filter (EKT) that incorporates diagonal length into the state vector, optimizes the motion model and noise parameters, and enables joint prediction of target position and morphology, thereby improving prediction accuracy.

State vector and model matrix design

By integrating the motion and morphological characteristics of mangoes, the Kalman filter is extended in terms of its state vector, measurement vector, and associated matrices, as detailed below:

- (1) **State vector:** An 8-dimensional vector $X = [x, y, w, h, L, dx, dy, dL]^T$ is adopted, where (x, y) are the target center coordinates, (w, h) the bounding-box width and height, L the diagonal length, and (dx, dy, dL) their respective velocities. Compared to the conventional 4-dimensional state vector, the newly added diagonal length and velocity terms provide a more comprehensive representation of the target's morphological and motion state, enhancing prediction robustness under varying shape conditions.
- (2) **Measurement vector:** A 5-dimensional vector $X = [x, y, w, h, L]^T$ is used, corresponding to the measured position and morphology values output by the instance extraction module, enabling effective observation of the state vector.
- (3) **Transition matrix:** Accounting for the cumulative effect of velocity terms, the transition matrix Φ is designed as follows, ensuring that the motion characteristics of the diagonal length are properly modeled and that position and morphology are predicted jointly.

$$\Phi = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (29)$$

- (4) **Observation matrix:** A 5×8 matrix H is employed, which retains only the dimensions corresponding to the measurement vector while

setting all other dimensions to zero, thereby achieving precise mapping from the state vector to the measurement vector.

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad (30)$$

- (5) **Noise matrices:** Noise parameters are optimized based on the motion characteristics of mangoes, balancing prediction smoothness and response speed. The process noise covariance is set to $Q = I_8 \times 0.003$ (where I_8 denotes the 8×8 identity matrix), and the measurement noise covariance is set to $R = I_5 \times 0.03$, reducing the impact of measurement errors on the filtering results.

Forecasting and update process

The core workflow of the Enhanced Kalman Filter comprises two stages, prediction and update, forming an inter-frame dynamic correction mechanism. (1) Prediction stage: Based on the historical tracking state and the transition matrix, the state prediction equations compute the prior state estimate $X(k|k-1)$ and prior error covariance $P(k|k-1)$ for the current frame, yielding the predicted bounding box, diagonal length, and corresponding motion velocities. (2) Update stage: When a matching target is detected in the current frame, the Kalman gain K_k is calculated using the measurement vector Z_k , and the state update equations correct the prior estimate to obtain the posterior state estimate $X(k|k)$ and posterior error covariance $P_{k|k}$. This dynamic correction of prediction errors enhances the accuracy of subsequent matching.

2.3.3. Multi-Feature Fusion Matching Module

Target association matching is the core component of the tracker, directly determining tracking accuracy. To address the failure of single IoU matching in scenarios involving occlusion and shape variations, a multi-feature fusion matching mechanism is designed. This mechanism constructs a comprehensive similarity matrix by weighting and fusing four types of feature similarities, and applies the Hungarian algorithm to achieve globally optimal matching between predicted targets and current instances.

Feature similarity calculation

For the four extracted features—position, shape, color, and diagonal length—corresponding similarity calculation methods are designed to leverage their complementary advantages, as detailed below:

- (1) **IoU Similarity (S_{IoU}):** Measures the spatial overlap between the predicted bounding box and the current detection bounding box, with a value range of $[0,1]$. A higher value indicates greater positional proximity. The calculation formula is:

where B_{pred} is the predicted bounding box, B_{det} is the current detection bounding box, and $Area(\cdot)$ denotes the bounding box area.

$$S_{IoU} = \frac{Area(B_{pred} \cap B_{det})}{Area(B_{pred} \cup B_{det})} \quad (31)$$

- (2) **Shape Similarity (S_{shape}):** Computed based on the cosine distance of Hu moment vectors. The cosine distance is scale-insensitive, enhancing the robustness of shape matching. The formula is:

$$S_{shape} = 1 - \frac{H_{pred} \cdot H_{det}}{\|H_{pred}\| \cdot \|H_{det}\|} \quad (32)$$

where H_{pred} and H_{det} are the Hu moment vectors of the predicted target and the current instance, respectively. $\cdot$ denotes the dot product, $\|\cdot\|$ denotes the L2 norm. S_{shape} ranges from $[0,1]$, with a higher value indicating greater shape similarity.

- (3) **Color Similarity (S_{color}):** Calculated using the Bhattacharyya distance between HSV histograms. This distance effectively measures the similarity between two probability distributions, making it suitable for color histogram matching and robust to illumination variations. The formula is:

$$S_{color} = 1 - \sqrt{1 - \frac{1}{\sqrt{\mu_{pred}\mu_{det}} \cdot N} \sum_{i=1}^N \sqrt{h_{pred}(i) \cdot h_{det}(i)}}} \quad (33)$$

where μ_{pred} and μ_{det} represent the means of the two histograms, respectively. $h_{pred}(i)$ and $h_{det}(i)$ are the frequencies of the i -th bin in the respective histograms, and N is the total number of bins. S_{color} ranges from $[0,1]$, with a higher value indicating closer color similarity.

- (4) **Diagonal Similarity (S_{diag}):** Combines category consistency and the ratio of diagonal lengths to ensure cross-category matching is invalid while leveraging the stability of diagonal length for reliable matching. The formula is:

$$S_{diag} = \frac{\min(L_{pred}, L_{det})}{\max(L_{pred}, L_{det})} \times \delta(c_{pred}, c_{det}) \quad (34)$$

where L_{pred} and L_{det} are the diagonal lengths of the predicted target and current instance, respectively, and $\delta(\cdot)$ is an indicator function that takes the value 1 when the categories are consistent, and 0 otherwise.

Weighted fusion and optimal matching

To emphasize the strong representational power of diagonal morphological features, the weighting strategy for four feature categories was optimized through extensive comparative experiments: $W_{IoU} = 0.25$, $W_{shape} = 0.25$, $W_{color} = 0.2$, $W_{diag} = 0.3$. The formula for calculating the composite similarity is:

$$S_{total} = W_{IoU} \cdot S_{IoU} + W_{shape} \cdot S_{shape} + W_{color} \cdot S_{color} + W_{diag} \cdot S_{diag} \quad (35)$$

Based on the comprehensive similarity scores, an $M \times N$ similarity matrix is constructed, where M is the number of predicted targets and N is the number of current instances. The matching problem is then formulated as a linear assignment problem. The Hungarian algorithm is employed to solve for the globally optimal matching, ensuring that the total matching cost between predicted targets and current instances is minimized. A similarity threshold of 0.5 is set; matches falling below this threshold are considered invalid, thereby avoiding tracking errors caused by incorrect associations.

2.3.4. ID Lifecycle management and stabilization module

ID switching is a core bottleneck that constrains the continuity of target tracking, particularly in complex scenarios such as temporary occlusion and missed detections between frames. To effectively mitigate this issue, this study designs an ID lifecycle management strategy, establishing a coordinated mechanism of caching, smoothing, destruction, and reuse. This approach enables dynamic control and stable output of ID states. Its core logic is illustrated in Fig. 9.

To balance tracking continuity and computational resource efficiency, this study, through empirical testing and iterative optimization, has defined four core control parameters. $ID_CACHE_FRAMES = 8$ sets the threshold for caching tentative IDs, providing an observation window for newly assigned or reused IDs to avoid erroneous deletion due to transient detection misses; this value can be adjusted in deployment

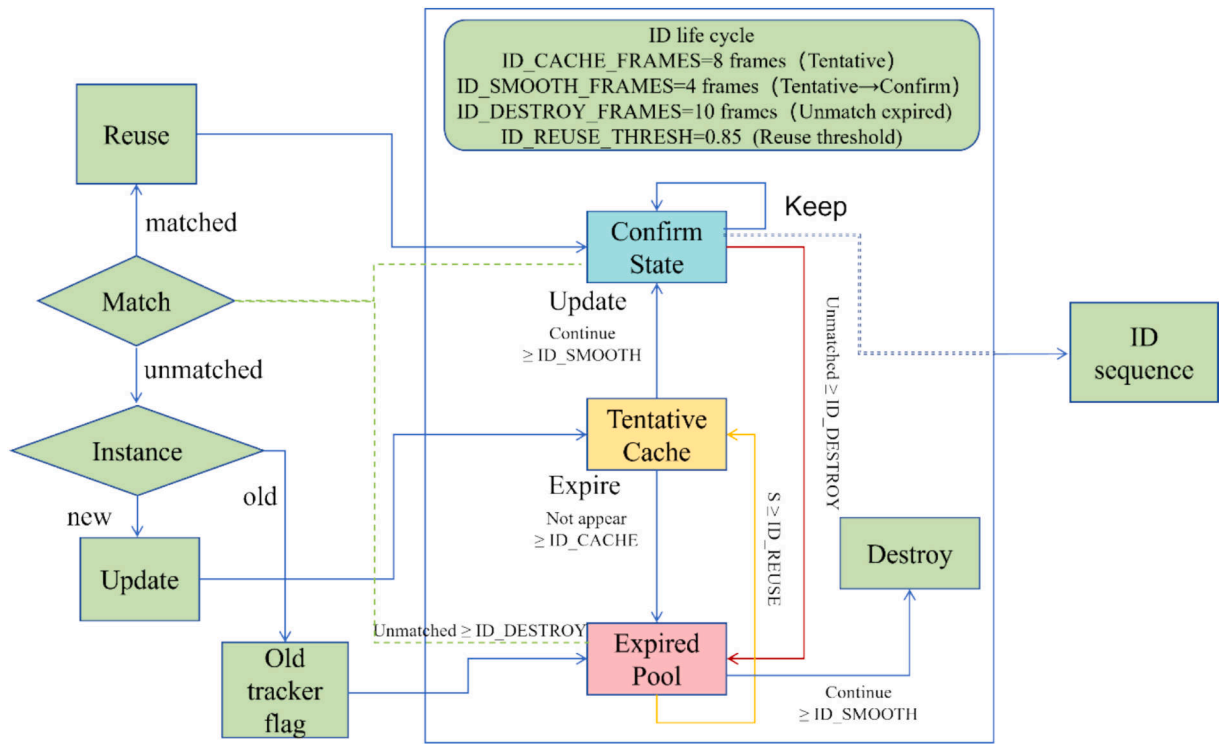


Fig. 9. ID Lifecycle Management Process Diagram.

based on the frequency of missed detections. $ID_SMOOTH_FRAMES = 4$ governs the transition from tentative to confirmed state, requiring an ID to be stably present for four consecutive frames before upgrading, thereby effectively filtering transient noise and accommodating scenarios with intense illumination changes. $ID_DESTROY_FRAMES = 10$ defines the destruction threshold, where IDs unmatched for ten consecutive frames are marked as expired and moved to an expired pool, preventing inactive IDs from consuming computational resources; this threshold can be dynamically tuned according to hardware capabilities. $ID_REUSE_THRESH = 0.85$ serves as the similarity criterion for reusing expired IDs; the threshold can be raised to reduce false reuse when fruit shapes are highly similar, or slightly lowered to increase reuse rates when shape variability is greater. Three ID states are defined: tentative, confirmed, and expired. The confirmed state enables continuous tracking and state updates, while the expired state follows rules for either reuse or permanent removal, forming a complete lifecycle.

Building on this parameter framework, the management process concurrently performs stabilization and adapts to different matching scenarios. When a match is successful, the historical ID is directly reused, and the Kalman filter state along with target features are updated to maintain tracking in the confirmed state. If the reused ID remains tentative, the smooth frame count is accumulated, and the ID is automatically upgraded once the threshold is met. In cases of no match, new instances are assigned a new ID and set to tentative, initiating a cache validation process. For historical tracking targets, the unmatched frame count is accumulated; if the destruction threshold is not reached, the ID is retained and its position is continuously predicted via the Kalman filter to ensure quick reconnection after occlusion, otherwise it is marked as expired. The process incorporates a dual stabilization mechanism. First, a smoothing strategy retrospectively examines the

tracking records of the previous four frames, prioritizing the historical ID that appears most frequently at the same spatial location and meets a similarity threshold, thereby suppressing transient ID switches; this similarity threshold can be flexibly adjusted based on fruit density. Second, the preset reuse threshold is used to traverse the expired pool, selecting high-similarity IDs for reuse. Reused IDs are reset to tentative for re-validation, which reduces new ID allocations at the source and enhances tracking continuity and stability.

Algorithm 1. Illustrates the ID matching process, which takes mango masks and previous ID sequences as inputs and outputs the current ID sequence. First, valid mango instances are extracted from the input segmentation mask. For each instance, four core features, namely IoU, shape, color, and diagonal length, are computed. A similarity matrix S is then constructed between the current instances and the previous frame's ID set. Matrix elements corresponding to instances of different categories are set to zero, while for instances of the same category, the four features are fused with assigned weights. Based on the cost matrix, the Hungarian algorithm is used to obtain the optimal matching pairs between current instances and historical IDs. Matching pairs with a similarity greater than 0.5 are directly assigned the corresponding historical ID, while unmatched instances are temporarily assigned $ID=None$. For instances without an ID, priority is given to reusing IDs from the expired ID pool that exhibit high similarity. If no suitable ID is available for reuse, a new ID is allocated. Simultaneously, the expired ID pool is updated by including IDs that failed to match in the previous frame, completing the closed-loop ID lifecycle management. Finally, the Kalman filter state is updated for instances that retain an existing ID, and a new Kalman filter is initialized for newly assigned IDs. The IDs of all instances are then organized and output as the current frame's ID

sequence, achieving continuous target tracking and stable ID assignment.

Algorithm 1 Multi-Feature Fusion and ID Lifecycle Management for Tracking and Counting (MFFT) Input: $Mask_{seg}$, IDs_{prev} , $IDs_{expired}$ Output: IDs_{curr}

```

1:  $Insts_{curr} = \text{ExtractInstances}(Mask_{seg})$ 
2: for  $Inst$  in  $Insts_{curr}$  do
3:  $Inst.feats = [IOU, Shape, Color, Diag]$ 
4: end for
5:  $S = \text{zeros}(|Insts_{curr}|, |IDs_{prev}|)$ 
6: for  $i \leftarrow 1$  to  $|Insts_{curr}|$  do
7: for  $j \leftarrow 1$  to  $|IDs_{prev}|$  do
8:  $S[i,j] = 0$  if  $Insts_{curr}[i].cls \neq IDs_{prev}[j].cls$  else  $w_1 \cdot IOU + w_2 \cdot Shape + w_3 \cdot Color + w_4 \cdot Diag$ 
9: end for
10: end for
11:  $matched = \text{Hungarian}(1 - S)$ 
12: for  $(i,j)$  in  $matched$  do
13:  $Insts_{curr}[i].id = IDs_{prev}[j].id$  if  $S[i,j] > 0.5$  else None
14: end for
15: for  $Inst$  in  $Insts_{curr}$  where  $Inst.id$  is None do
16:  $Inst.id = \text{ReuseID}(IDs_{expired}, Inst.feats)$  or  $\text{NewID}()$ 
17: end for
18:  $IDs_{expired} = IDs_{prev} \setminus [Inst.id \text{ for } Inst \text{ in } Insts_{curr}]$ 
19: for  $Inst$  in  $Insts_{curr}$  do
20:  $\text{UpdateKalman}(Inst)$  if  $Inst.id$  exists else  $\text{InitKalman}(Inst)$ 
21: end for
22:  $IDs_{curr} = [Inst.id \text{ for } Inst \text{ in } Insts_{curr}]$ 

```

Following the construction of the tracker, our method was deployed in mango orchards across Hainan Province. As illustrated in Fig. 10, a representative example is provided by extracting frames at intervals of two from a randomly selected tracking video. Five sequential images are presented to demonstrate the segmentation and tracking process. Mangoes on trees are delineated with blue contours and red diagonal lines. Corresponding blue text boxes positioned above these mangoes sequentially display the ID, mango variety, and diagonal pixel length. Conversely, mangoes on the ground are outlined with green contours and red diagonal lines, with green text boxes indicating their respective ID, variety, and diagonal pixel length. To explicitly visualize the ID evolution, light blue bounding boxes denote the continuous tracking of an individual mango with a persistent ID. Red bounding boxes signify the detection of a new mango instance, which is assigned a novel ID. Finally, black bounding boxes indicate the disappearance of a mango from the frame, resulting in the termination of segmentation and the expiration of its ID. Our method effectively accomplishes tracking and ID assignment tasks. The designed ID lifecycle mechanism facilitates the extension of tracking longevity for individual mango instances while maintaining rapid responsiveness to new and existing entities. However, tracking-by-counting still faces a significant challenge regarding ID switches. In mango orchards characterized by variable shooting angles and severe occlusion, the disruption of motion trajectories and the

impact of angle-dependent illumination on target appearance render ID switches inevitable.

To visualize this issue, a video was randomly selected and processed (Fig. 17). The ByteTrack tracker was applied to the same video, and corresponding frames were extracted for comparison. The total counts of two mango categories were analyzed. As illustrated in the enlarged cropped images, ID switching is a prevalent issue. Variations in segmentation performance across different experimental runs are considered normal occurrences. Through comparison, our method assigned only 24 distinct IDs within the same timeframe, whereas ByteTrack generated 28 distinct IDs. This demonstrates the effectiveness and feasibility of our approach in mitigating the ID switch problem.

2.4. Experimental Setup and Evaluation Metrics

2.4.1. Hardware and software

All experiments in this study were conducted on a Windows 11 terminal equipped with an AMD Ryzen 7 5800H processor and an NVIDIA GeForce RTX 3060 Laptop GPU (6 GB VRAM). Model training, inference, and computations were performed on the GPU, supported by the NVIDIA CUDA 11.8 toolkit. The software environment was built on Python 3.10.19, with deep-learning model construction, training, and inference implemented using the PyTorch 2.7.1 framework and the TorchVision 0.22.1 library. Image preprocessing and feature extraction relied on OpenCV 4.13.0, while numerical computation and data preprocessing were supported by NumPy 2.2.6 and Scikit-learn 1.7.2.

2.4.2. Training hyperparameters

During the training phase of the mango segmentation model, a suitable training configuration was established through experimental validation. To accommodate the 6 GB VRAM of the laptop GPU, the input image size was uniformly set to 256×256 , and the batch size was fixed at 1. An 8-step gradient accumulation strategy was employed, resulting in an effective batch size of 8, which ensured stable gradient updates while avoiding VRAM overflow. A hierarchical learning-rate strategy was adopted: the global initial learning rate was set to 0.0001, with lower rates assigned to the backbone (0.000005 for Stage 1–2, 0.00002 for Stage 3–4) to preserve pre-trained features, and the decoder kept at 0.0001 to accelerate adaptation to mango-specific characteristics. The model was trained for 100 epochs using the AdamW optimizer (weight decay 0.0001, $\beta_1 = 0.9$, $\beta_2 = 0.999$) to mitigate overfitting and smooth gradient updates for the imbalanced sample distribution. Given the small training set, a combined warm-up and cosine-annealing learning-rate schedule was applied to prevent overfitting. A 3-epoch linear warm-up (start_factor = 0.01) was followed by cosine annealing over the remaining 97 epochs ($T_{max} = 97$, $\eta_{min} = 1e-7$), allowing fine-tuning in later stages. Mixed-precision training (AMP) was enabled to improve efficiency and reduce memory consumption, and an

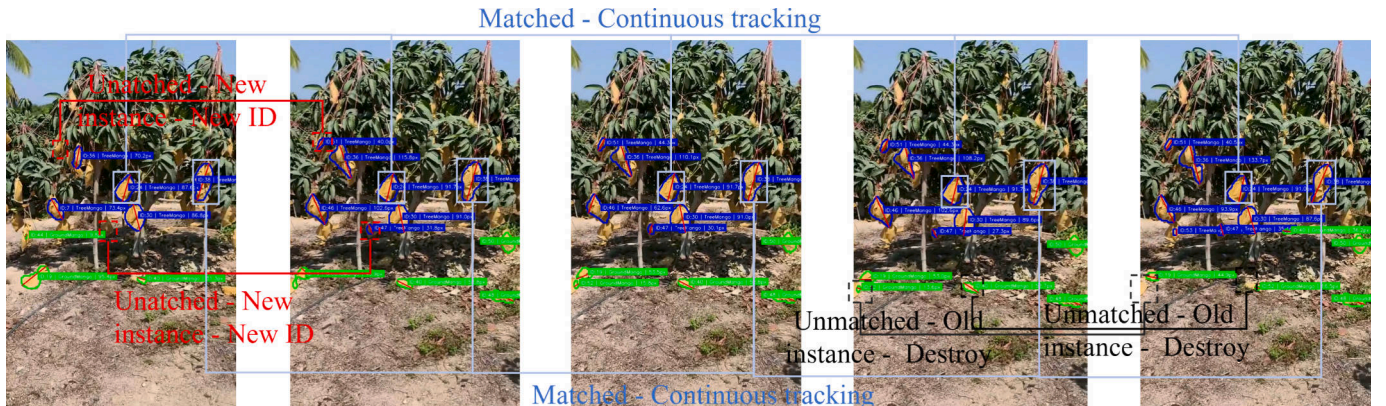


Fig. 10. Randomly selected output video as an example of segmentation tracking results.

early-stopping rule terminated training if validation metrics did not improve for 20 consecutive epochs, balancing convergence with computational efficiency.

2.4.3. Evaluation indicators

The following metrics are employed to evaluate network performance. The assessment encompasses two categories: segmentation accuracy and model efficiency. For segmentation accuracy, precision and recall measure the probabilities of false positives and false negatives, respectively:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (36)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (37)$$

where TP is true positives (correctly classified mango samples). FP is false positives (background or other classes misclassified as mango). FN is false negatives (mango samples misclassified as background or other classes).

Mean Intersection over Union (mIoU) serves as a core metric reflecting the overlap between predicted and ground-truth masks;

$$\text{mIoU} = \frac{1}{k+1} \sum_{i=0}^k \frac{TP_i}{TP_i + FP_i + FN_i} \quad (38)$$

where k is the number of classes ($k = 1$ in this study, distinguishing mango and background). TP_i , FP_i , and FN_i represent the number of true positives, false positives, and false negatives in class i , respectively.

The F1-score balances the trade-off between precision and recall;

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (39)$$

Mask average precision and its derivatives (mAP50, mAP50-95) provide a comprehensive evaluation of both localization and mask-generation accuracy in instance segmentation.

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}(i) \quad (40)$$

where N is the number of object classes; AP is the average precision of class i .

Regarding model efficiency, the number of parameters reflects model complexity and memory consumption, floating-point operations (FLOPs) quantify computational overhead, and inference frames per second (FPS) along with average inference time directly indicate real-time processing capability, ensuring suitability for mobile orchard deployment.

For the mango tracking task, four core metrics are selected for performance assessment. The IDF1 score balances the accuracy and completeness of target-ID matching, suppressing ID switches:

$$\text{IDF1} = 2 \times \frac{\text{IDTP}}{\text{IDTP} + \text{IDFP} + \text{IDFN}} \quad (41)$$

where, IDTP is correctly matched IDS, IDFP is mismatched IDS, IDFN is unmatched GT IDS.

multiple object tracking accuracy (MOTA) integrates false-positive, false-negative, and ID-switch errors, serving as a key indicator of overall tracking performance:

$$\text{MOTA} = 1 - \frac{\text{FP} + \text{FN} + \text{IDS}}{\text{GT}} \quad (42)$$

where IDS is ID switches, GT is total ground-truth objects.

Multiple object tracking precision (MOTP), based on bounding-box overlap, quantifies the accuracy of location prediction;

$$\text{MOTP} = \frac{1}{\text{TP}_{\text{total}}} \sum_{t,i} \text{IoU}(t,i) \quad (43)$$

where TP_{total} is total true positives across all frames, $\text{IoU}(t,i)$ is IoU between detection and ground truth in frame t for object i .

Tracking inference time evaluates the real-time performance of the tracking module, ensuring efficient operation of the full pipeline.

3. Results and discussions

3.1. Segmentation results

3.1.1. Model performance

To validate model performance, comparisons with mainstream state-of-the-art methods are essential. Experiments were conducted on the same dataset utilizing Mask R-CNN, Mask2Former, DETR, and YOLOv8 s, all of which are widely adopted advanced methods for crop segmentation and recognition. The experimental results for various metrics are presented in Table 1. Our method demonstrates outstanding comprehensive performance. It achieves the highest Precision of 0.82 among all models, indicating a lower false detection rate, while also attaining an excellent Recall of 0.78, which validates its capability to effectively identify the vast majority of mango targets. Our mAP50 and mAP50_95 scores are 0.67 and 0.33, respectively. These values are close to the scores of 0.69 and 0.36 achieved by Mask R-CNN, suggesting performance approaching that of mature instance segmentation models. Furthermore, our method comprehensively surpasses Mask2Former in mIoU and F1 score, reaching 0.71 and 0.73 respectively, which reflects high pixel-level accuracy. Although DETR and YOLOv8 s exhibit higher mAP metrics, with mAP50 scores of 0.78 and 0.84 respectively, and superior recall rates—attesting to their robust instance segmentation capabilities—our method offers greater practical value by balancing segmentation accuracy with inference speed. With an inference time of 25.67 milliseconds, our approach is significantly faster than mainstream models such as YOLOv8 s and DETR. Notably, this model prioritizes pixel-level segmentation accuracy in complex occluded orchard scenarios, focusing on boundary and small-object feature extraction rather than maximizing detection mAP. Thus, its mAP50 and mAP50_95 are lower than YOLOv8 s and DETR. To ensure segmentation robustness, we use the Swin-T backbone and multi-attention modules, resulting in slightly higher parameters and computation, leading to inferior performance in some localization metrics. Therefore, enhancing segmentation performance remains the primary focus of our future research Table 2.

We analyze the mechanisms behind the metric differences. YOLOv8 and DETR achieve higher detection mAP. However, our method obtains better mIoU and F1 scores. The root cause is that our BAM effectively enhances edge responses in adhesive regions. The background gating module suppresses the weights of interfering features such as soil and weeds. YOLO-style models focus on bounding box classification. They lack sufficient modeling of boundary pixels for adhesive mangoes. This leads to missed detections of small occluded fruits. DETR tends to over-predict and generate redundant masks. The pixel-level segmentation defects of both model types directly cause counting errors. Our model achieves faster inference speed than Mask R-CNN and DETR. This benefits from the hierarchical feature extraction efficiency of the Swin-T backbone and the lightweight design of our custom decoder. These features ensure accuracy while avoiding redundant computations.

To evaluate the performance of the segmentation network, validation visualization images were generated at each epoch during the model verification phase. Representative images from diverse environments are presented here. Mangoes on trees are overlaid with blue masks, and corresponding blue text boxes positioned above display the ID, mango variety, and diagonal pixel length. Conversely, mangoes on the ground are overlaid with green masks, with green text boxes indicating their ID and variety. Red lines delineate the diagonal of each

Table 1

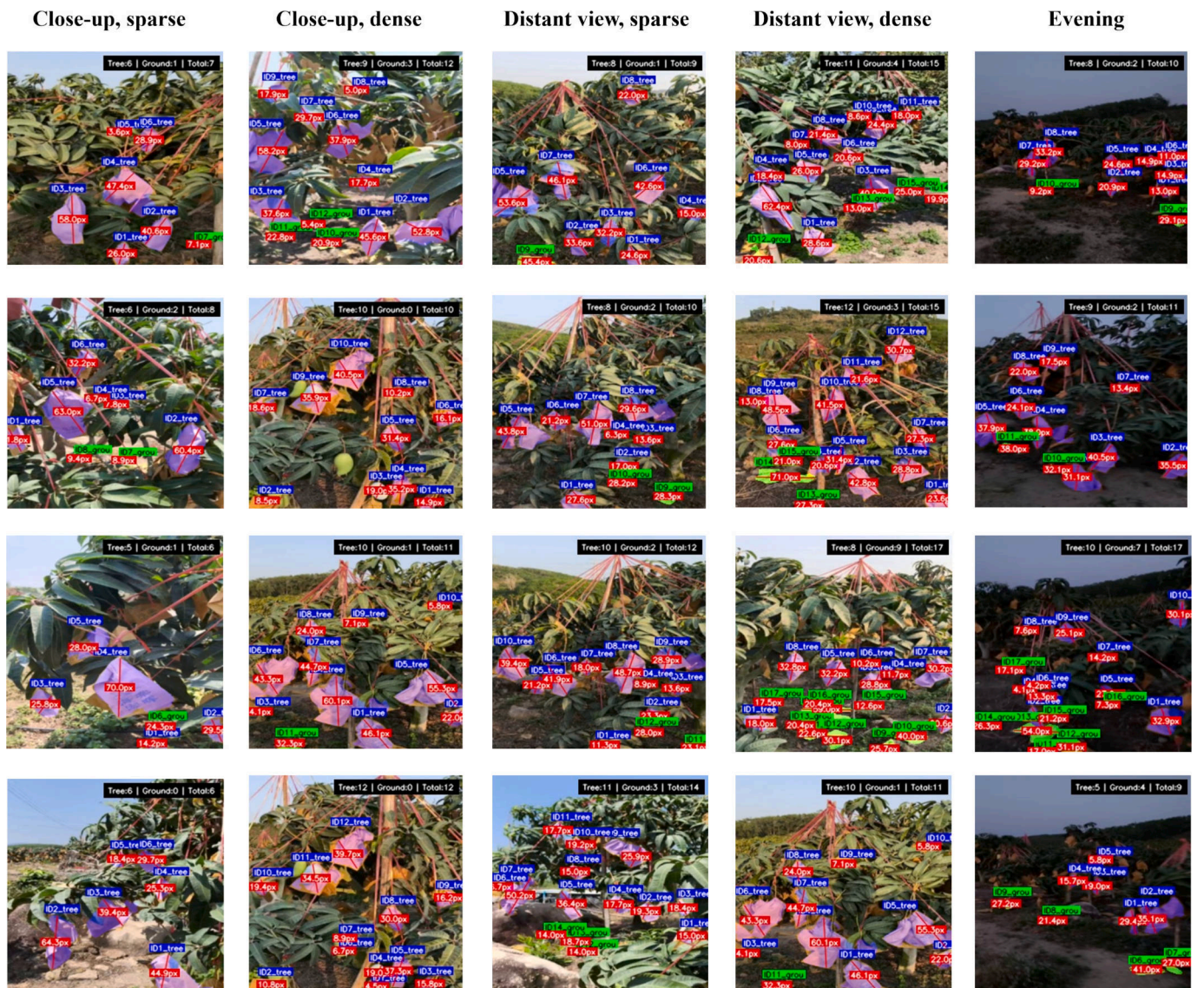
Comparison of segmentation results based on different advanced methods.

Model	Precision	Recall	mIoU	F1	mAP50	mAP50_95	Param	FLOPs	FPS	Infer_time(ms)
MaskRCNN	0.78	0.80	-	-	0.69	0.36	43.93	401.74	6.28	159.34
Mask2former	0.72	0.77	0.64	0.67	0.63	0.30	54.54	20.58	10.36	96.50
DETR	0.75	0.85	-	-	0.78	0.40	36.96	9.53	21.77	45.94
YOLOv8s	0.79	0.81	-	-	0.84	0.47	11.8211	20.172	24.88	41.40
Ours	0.82	0.78	0.71	0.73	0.67	0.33	33.77	41.86	40.32	25.67

Table 2

Ablation experiment results based on attention modules.

	mIoU	F1	mAP50	mAP50_95	Param (M)	FLOPs (G)	FPS	Infer_time(ms)
No Attention	0.64	0.65	0.55	0.23	31.04	28.98	41.43	23.88
Attention-Deficit Module	0.69	0.71	0.61	0.29	32.01	27.62	40.33	24.80
Attention CBAM	0.67	0.68	0.57	0.25	31.32	43.37	48.36	22.40
Attention SE	0.64	0.67	0.59	0.27	33.76	41.83	44.51	22.50
Ours	0.71	0.73	0.67	0.33	33.77	41.86	40.32	25.67

**Fig. 11.** Visualization Results of the Segmentation Network.

mango, and the pixel length is annotated above these lines. Note that the IDs are assigned per image. The top-right corner of each image summarizes the count of mangoes per category and the total count. Selected

representative environmental scenarios from the validation set are visualized in Fig. 11 to demonstrate the classification and segmentation efficacy. From left to right, the scenarios include close-up sparse

distribution, close-range dense distribution, long-range sparse distribution, long-range dense distribution, and low-light conditions during dusk. These visualizations collectively demonstrate that our network achieves superior segmentation performance across various environmental conditions.

To provide a more intuitive reflection of the regions of interest within the images processed by the neural network, the Grad-CAM method was employed for analysis. In the Grad-CAM visualizations, red regions indicate the highest level of network attention, followed by green, while blue denotes areas of low attention. Superior recognition performance is indicated if the red regions accurately overlay the mango targets. As illustrated in Fig. 12, our method demonstrates precise focus

on the mangoes, with red activations concentrated within the target regions and minimal interference from the background. Among the comparative methods, DETR also exhibits accurate attention allocation but demonstrates insufficient focus on certain ground mangoes. YOLOv8s displays relatively limited attention regions. In contrast, Mask R-CNN and Mask2Former are susceptible to background interference, characterized by extensive green activation areas.

3.1.2. Ablation experiment

To validate the efficacy of the proposed attention module, ablation studies were conducted by modifying the attention mechanism. Four network configurations were trained and evaluated: a baseline network

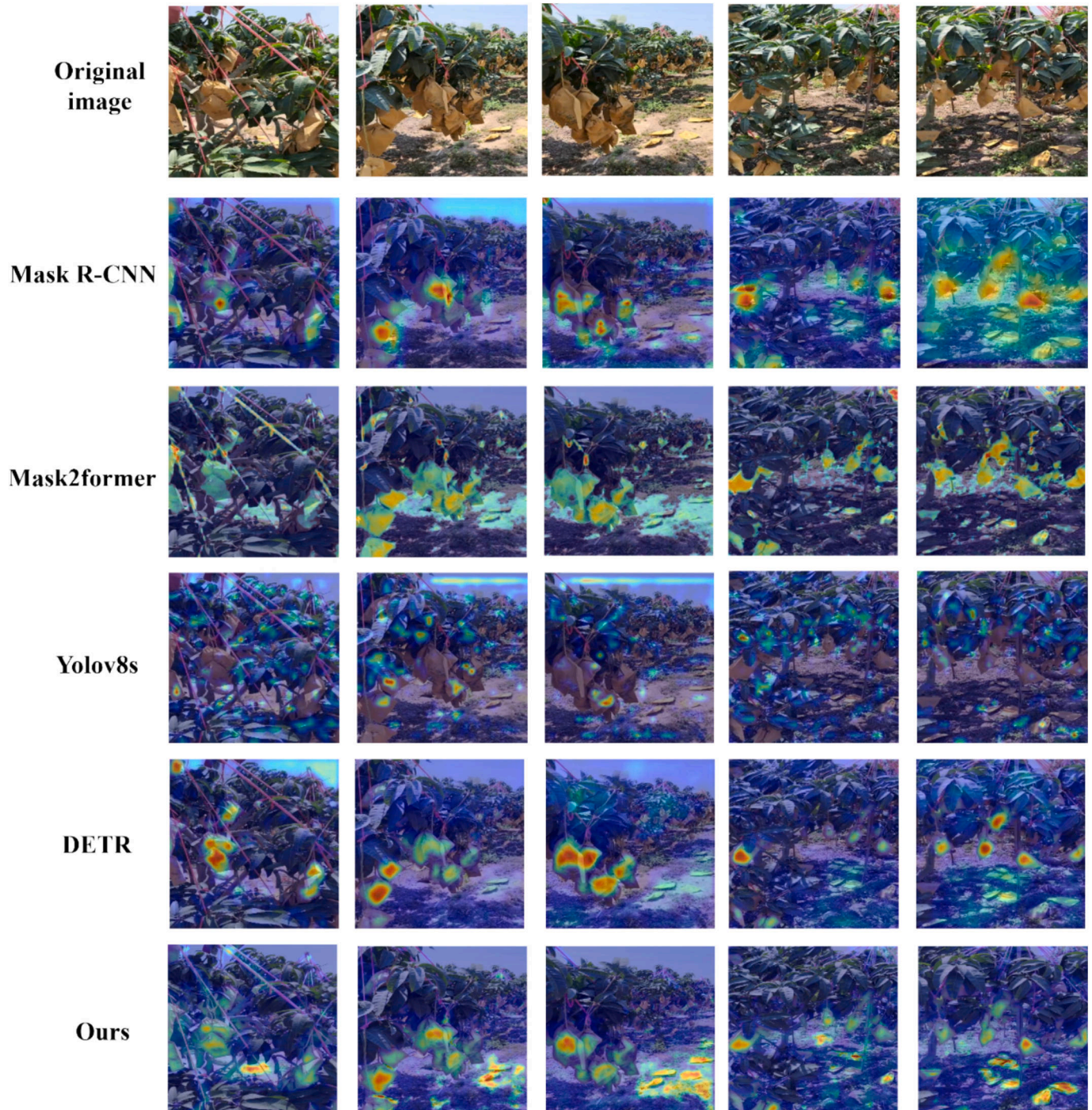


Fig. 12. Grad-CAM Visualization Results for Different Advanced Methods.

without any attention mechanisms, a network with the SOA, BAM, and BG modules removed (located between feature fusion and mask prediction), and networks where the attention module was replaced with the popular CBAM and SE modules.

The results demonstrate that the network incorporating our designed attention module achieves the highest mIoU and mAP50 scores of 0.71 and 0.73, respectively. Despite the increased computational complexity, this configuration yields the most superior segmentation performance. The network with reduced modules follows, achieving an mIoU of 0.69

and an mAP50 of 0.61, while the network without attention performs the poorest. Although the SE and CBAM modules are not specifically tailored to our scenario, they still yield performance improvements. The SE method achieves an mAP50 of 0.59, and the CBAM method achieves an mIoU of 0.67, reflecting their general applicability.

The Grad-CAM method was similarly employed to visualize the ablation experiments of the attention module. As illustrated in the Fig. 13, the network without attention mechanisms suffers from severe background interference, focusing extensively on background regions.

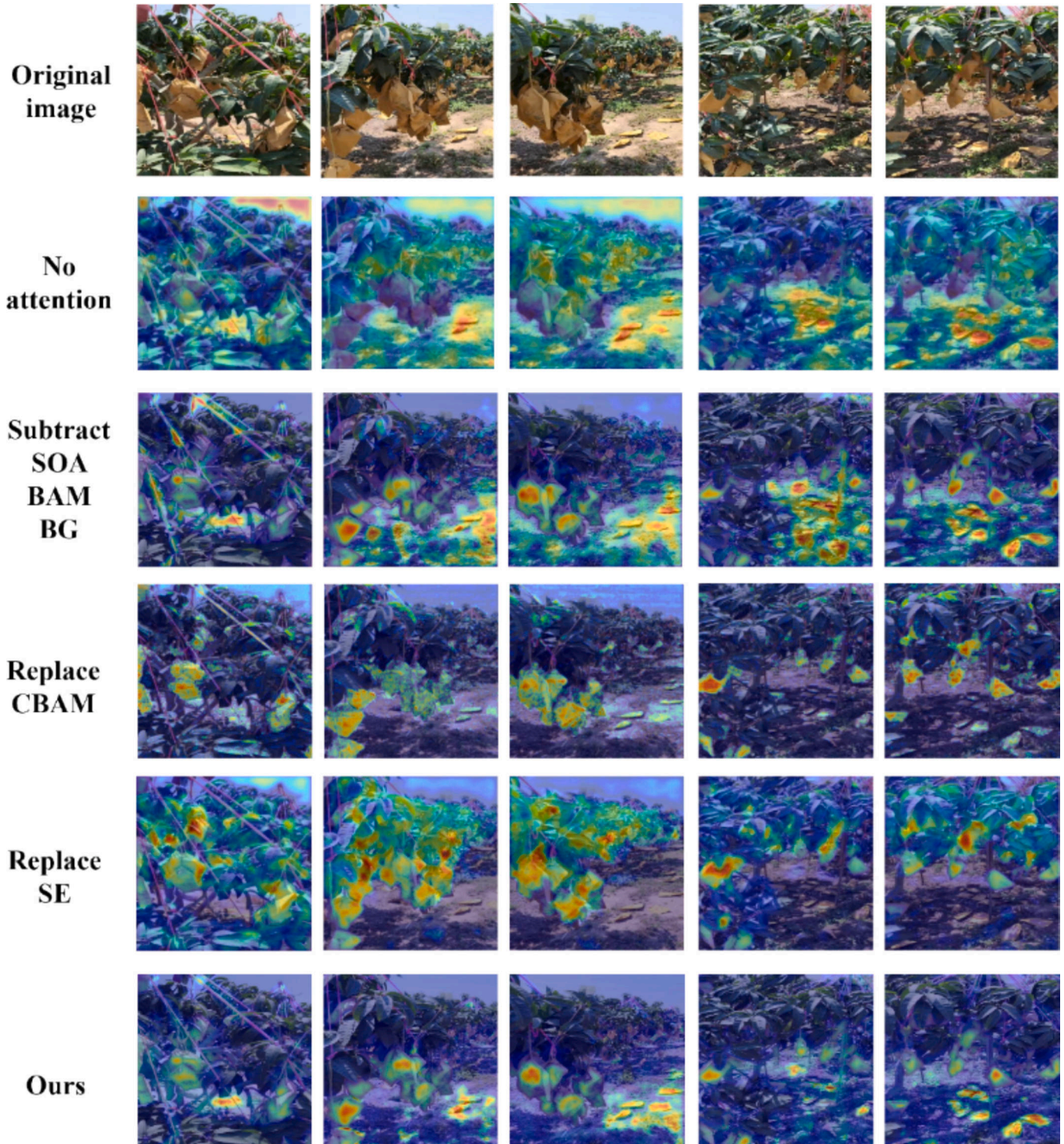


Fig. 13. Grad-CAM Visualization Results for Different Attention Modules.

The network with only the SOA, BAM, and BG modules removed successfully focuses on mango regions but remains susceptible to interference from ground areas. The network utilizing the SE attention module also focuses on a significant number of leaves. While the network utilizing the CBAM attention module demonstrates good focus, the red activation areas within the mango regions are sparse, indicating lower attention compared to our method. These results collectively demonstrate that the attention module designed in our method plays a significant role in enhancing recognition capabilities.

The selection of the backbone network is crucial for model performance. To validate the correctness of choosing the Swin Transformer Tiny (Swin-T) backbone, we conducted comparative experiments by replacing the backbone. We selected the high-performance and widely used CNN backbones EfficientNet-B0, EfficientNetV2-S, MobileNet_v3_large, and ConvNeXt_T for substitution. A total of four networks were trained and evaluated. The results, as presented in Table 3, demonstrate that our method achieves the highest segmentation metrics, including MIoU and mAP50, indicating superior segmentation performance. Other networks with alternative backbones exhibit advantages in terms of efficiency. Specifically, EfficientNet-B0 achieves an inference time of only 6.14 milliseconds. Additionally, ConvNeXt_T demonstrates favorable performance with an mAP50 of 0.61 and an inference time of 10.98 milliseconds. However, a significant gap remains between the segmentation metrics of these alternatives and our proposed method.

The Grad-CAM method was utilized to visualize the backbone ablation experiments. As illustrated in Fig. 14, the EfficientNet-B0 network exhibits high attention to mango regions but suffers from significant interference from soil and foliage. The EfficientNetV2-S network demonstrates concentrated activations on mangoes and reduces interference from distant backgrounds, yet interference in close-range scenes is exacerbated. Although MobileNet_v3_large and ConvNeXt_T achieve substantial coverage of mango regions with high attention, they still manifest considerable background activation and erroneous attention allocation. In contrast, our method effectively suppresses interference, maintaining focus primarily on the mango targets. These results substantiate that the integration of the Swin-T backbone within our method constitutes the optimal architectural choice.

Ablation experiments were subsequently conducted on the decoder, mask head, and boundary refinement module. As the decoder and mask head serve as essential basic components for segmentation and cannot be directly removed, we performed comparisons by replacing them with same-level alternative structures. For the decoder, the baseline adopted a simple upsampling decoder that only uses deconvolution for feature reconstruction, which is structurally simple and lacks multi-scale feature interaction and fusion capabilities. For the mask head, the baseline used a lightweight original mask head that only performs mask prediction through basic convolutions, lacking boundary constraints and scene adaptability. The boundary refinement module was validated by a controlled variable strategy (direct embedding or removal) to verify its refinement effect on blurry boundaries between fruits, branches, leaves, and weeds. The ablation results are presented in Table S1.

It can be observed that segmentation performance improves steadily as the decoder, mask head, and refinement module are gradually incorporated. The baseline model with a simple pixel decoder, basic mask head, and no refinement module achieves an MIoU of 0.67, F1 of

0.69, mAP50 of 0.63, with 32.17 M parameters and 41.22 FPS. After replacing it with the custom Transformer decoder, all metrics are significantly improved. Adding the improved mask head further enhances precision. The final model, equipped with the custom decoder, dedicated mask head, and boundary refinement module, achieves the best segmentation performance with an MIoU of 0.71, F1 of 0.73, and mAP50 of 0.67. Although the number of parameters increases to 33.77 M, the inference speed remains 40.32 FPS with only a slight drop, achieving a favorable balance between accuracy and efficiency. The results validate the synergistic benefits of the proposed modules.

To facilitate a deeper understanding of the network operational principles and enhance the interpretability of the deep learning mechanism, feature maps from various layers of the model were selectively visualized. These feature maps are presented as heatmaps to provide an intuitive representation of which regions are prioritized during propagation and to observe the efficacy of each module. As illustrated in Fig. 15, the visualization includes the original input image, the Grad-CAM output, and feature maps from different stages. Feature maps from layers 0 through 7 of the Swin-T backbone were selected to comprehensively demonstrate the feature evolution throughout the backbone, encompassing processes such as convolution, normalization, transformation, and patch merging. Subsequent visualization of the coordinate attention within the SFE and feature projection modules, along with the feature fusion heatmaps, reveals a progressive refinement of the captured features.

The SOA heatmap illustrates the prioritized capture of small target features. The BAM heatmap exhibits distinct mango contours, demonstrating robust boundary feature extraction capabilities. The Mask Head layer successfully extracts complete mango features, while the MaskRefineModule layer further refines these features to achieve a complete separation between the mango targets and the background.

Notably, perspective and shooting distance change continuously during orchard video collection, leading to significant appearance variations of the same mango across consecutive frames. To objectively evaluate the accuracy of the PCA-based diagonal calculation, this study randomly selected 240 valid and independent mango samples from test videos, covering diverse growth postures. Samples were divided into four groups according to occlusion levels: no occlusion, slight occlusion, moderate occlusion, and severe occlusion, so as to quantitatively analyze the effects of posture, perspective, and occlusion on measurement accuracy. Taking manually annotated diagonals as ground truth, the average relative error of each group was calculated, and the results are presented in Table S2.

It is evident that, after eliminating the interference of fruit contour adhesion, perspective switching, shooting distance, and posture deflection have little influence on PCA principal direction fitting, with only a limited increase in error. The main source of error is contour incompleteness caused by branch and leaf occlusion, which induces small deviations in the external diagonal. Overall, the proposed PCA-based diagonal method can adapt to dynamic orchard environments and maintains stable accuracy under regular conditions, providing reliable geometric features for fruit size statistics and yield estimation.

3.2. Tracking Count Results

To evaluate the efficacy of our approach, this study conducted a

Table 3
Ablation experiment results based on different backbone networks.

	MIoU	F1	mAP50	mAP50_95	Param (M)	FLOPs (G)	FPS	Infer_time(ms)
EfficientNet-B0	0.68	0.69	0.60	0.29	8.34	28.32	162.77	6.14
EfficientNetV2-S	0.67	0.68	0.59	0.26	24.88	113.29	37.83	26.44
MobileNet_v3_large	0.65	0.66	0.57	0.25	8.89	109.12	65.41	15.29
ConvNeXt_T	0.68	0.69	0.61	0.28	8.34	28.32	91.05	10.98
Ours	0.71	0.73	0.67	0.33	33.77	41.86	40.32	25.67

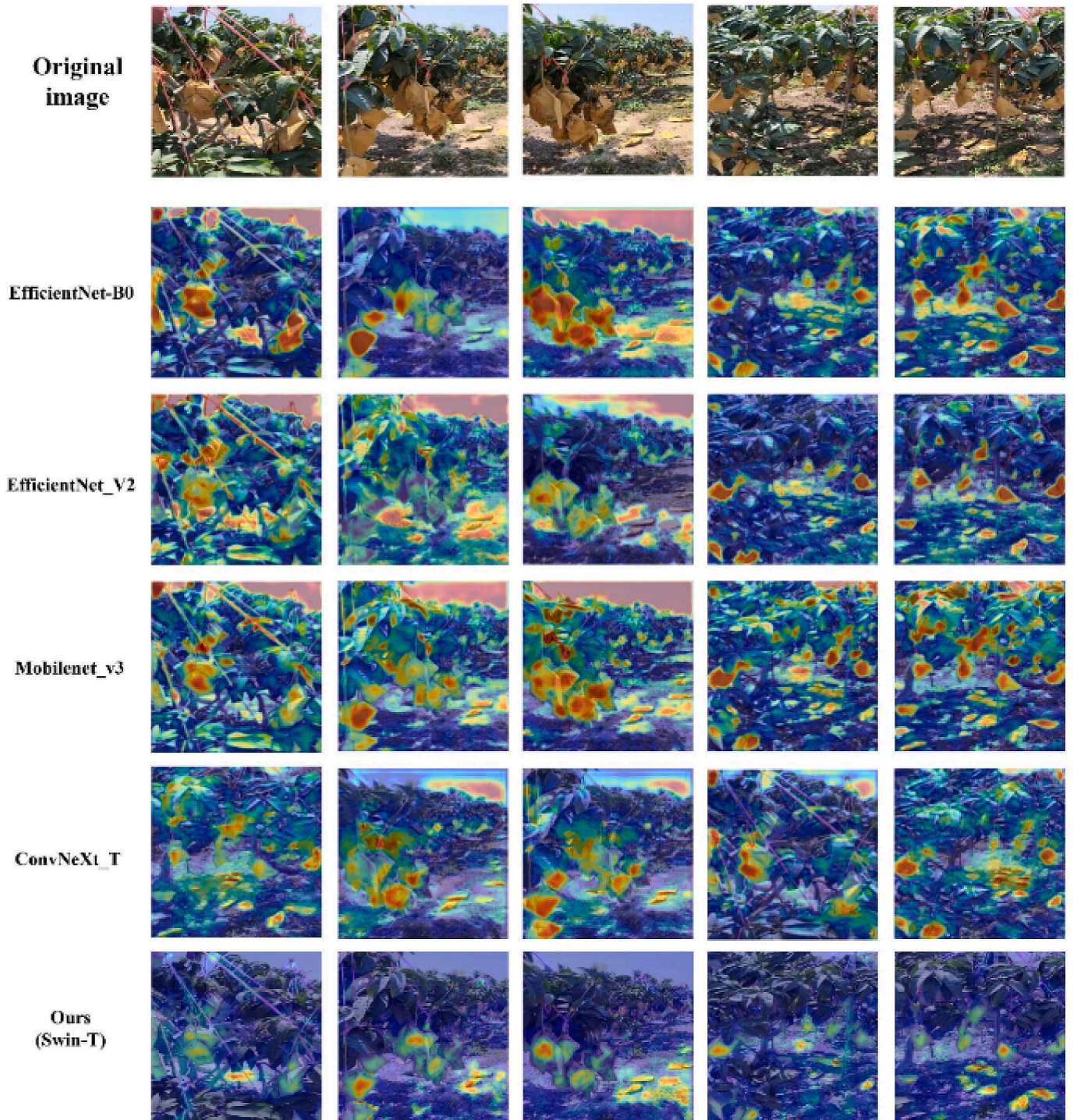


Fig. 14. Grad-CAM Visualization Results for Different Attention Modules.

systematic performance comparison and assessment of the proposed Multi-Feature Fusion Tracking algorithm against three current mainstream tracking algorithms, namely ByteTrack, DeepSORT, and StrongSORT, on the tasks of mango counting and biomass estimation in orchards. Our comparison is based on video data from forty data points. The results are presented in the form of scatter plots and comprehensive performance quantification tables as follows.

In the counting task, the regression equation of the MFFT algorithm exhibits a slope of 1.04, which is the closest to the ideal y equals x baseline. Combined with a goodness of fit of R squared equals 0.98, this indicates a high degree of agreement between the counting results and

the ground truth values. In contrast, the regression slopes of ByteTrack, DeepSORT, and StrongSORT are 0.65, 0.55, and 0.50 respectively, deviating significantly from 1 and exhibiting large intercept terms. This reflects a systematic bias of underestimation prevalent in these algorithms. In terms of error metrics, the RMSE, MAE, and MRE of MFFT are 12.95, 8.75, and 4.30 percent respectively, all of which are substantially lower than those of other algorithms. Notably, the MRE of StrongSORT is as high as 74.39 percent, further highlighting the superiority of MFFT in counting accuracy. The scatter distribution in Fig. 16 intuitively shows that MFFT consistently adheres closely to the y equals x diagonal, whereas other algorithms gradually deviate from the baseline as the true

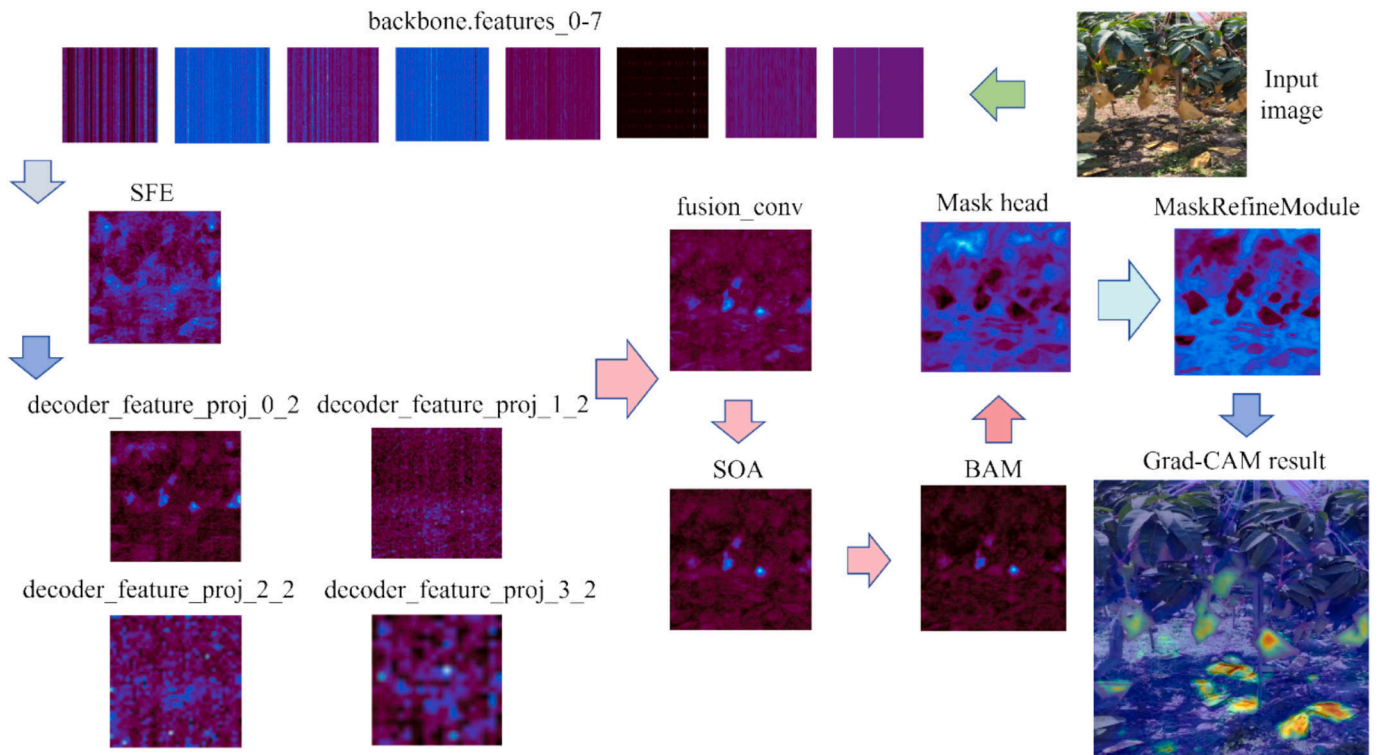


Fig. 15. Visualization results of feature maps from different modules in the network.

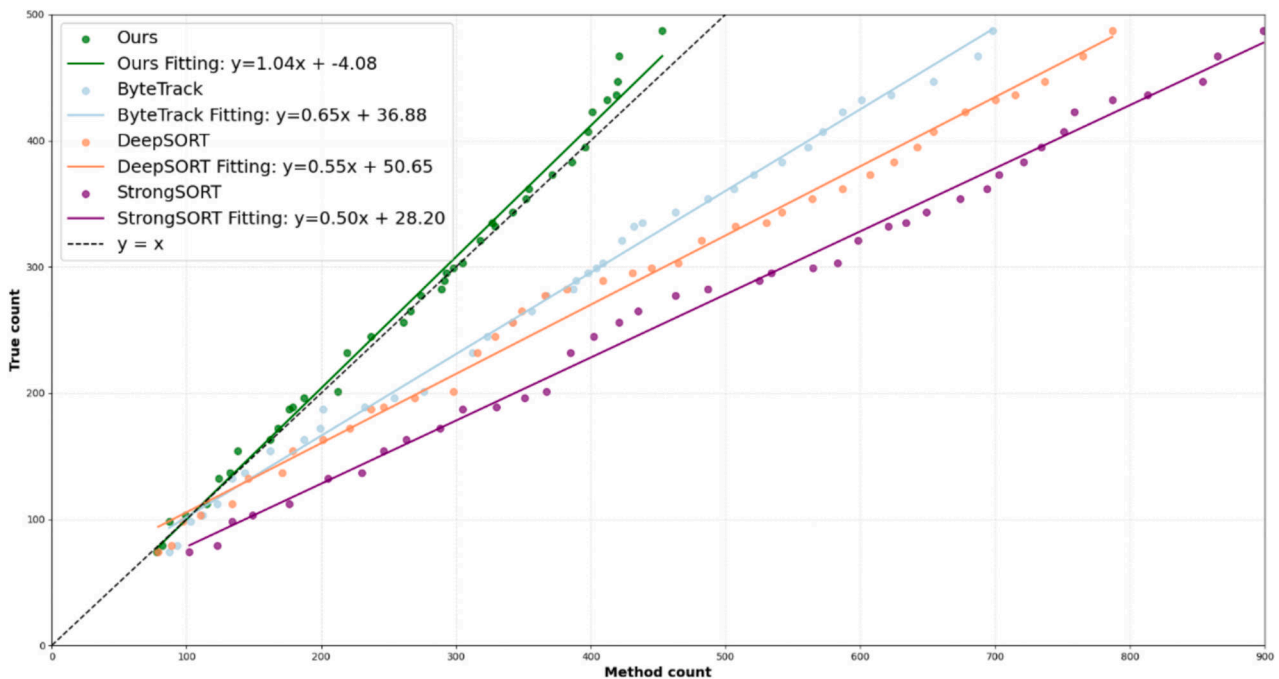


Fig. 16. A visual comparison of our method's counting results with those of three other popular tracking algorithms.

values increase, validating the analytical conclusions.

In the biomass estimation task, Fig. 19 and Table 4 present the results of biomass prediction based on diagonal length. MFFT demonstrates optimal performance with a regression equation of y equals $8.28x$ minus 3.24 . A goodness of fit of R squared equals 0.99 and an MRE of only 4.49 percent indicate extremely high reliability in predicting mango biomass. Compared to other algorithms, the regression slopes of ByteTrack, DeepSORT, and StrongSORT are 4.53 , 3.62 , and 3.02 respectively. These

slopes are not only much lower than that of MFFT but also accompanied by larger intercept terms, leading to overall lower biomass estimates. The visualization results in Fig. 19 further corroborate this point. The fitting line of MFFT maintains the closest linear relationship with the true values, whereas the fitting lines of other algorithms gradually move away from the baseline as the diagonal length increases, showing a distinct error accumulation effect. Furthermore, the RMSE and MAE of MFFT in biomass estimation are 6.09 and 4.79 respectively, significantly

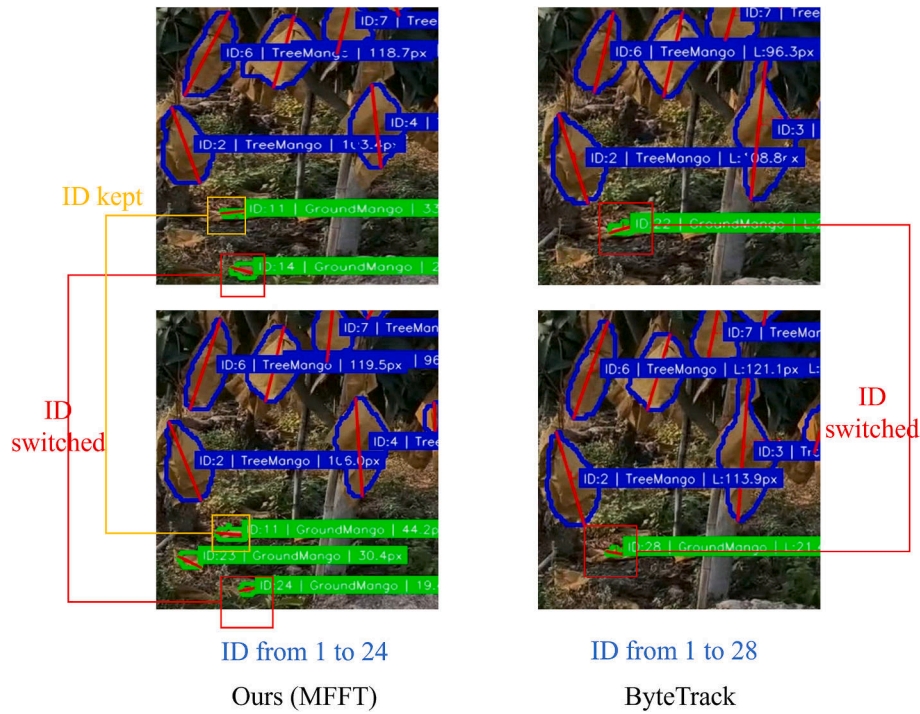


Fig. 17. Comparison of Our Method with ByteTrack's Visualization Results.

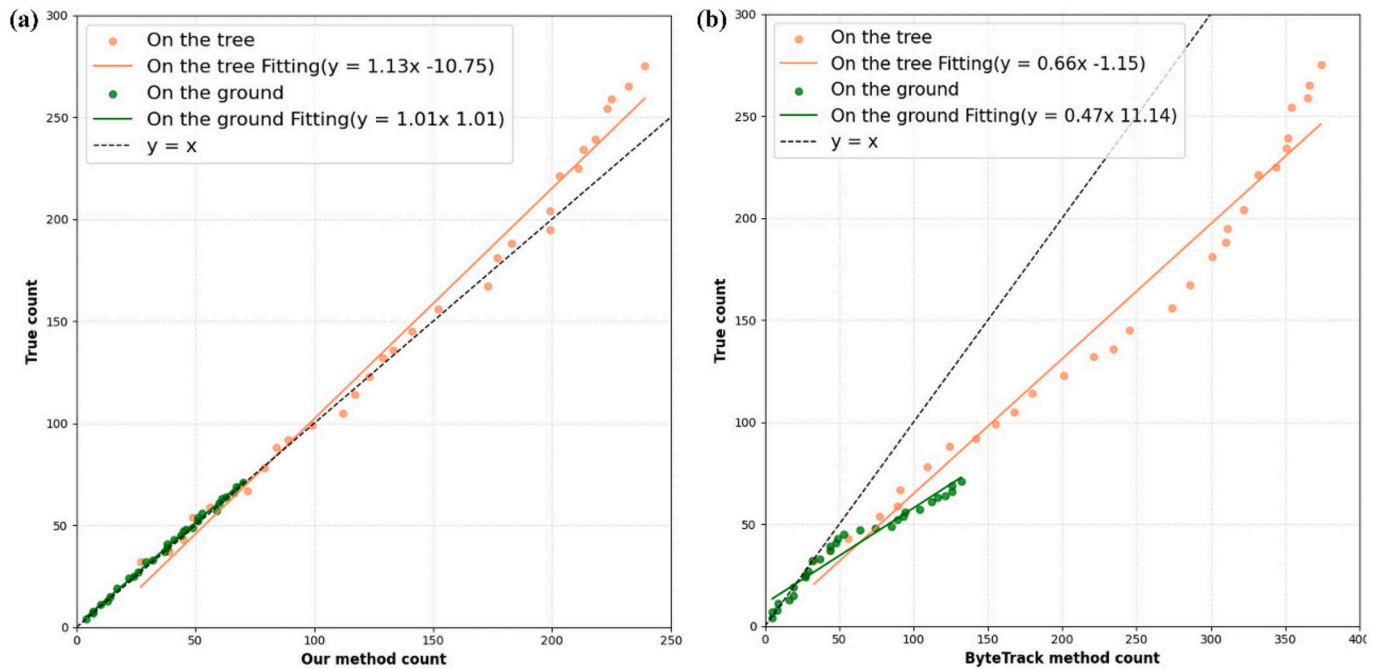


Fig. 18. Visualization of data points in Table 5 (a) MFFT (our method) (b) ByteTrack.

outperforming other algorithms. This demonstrates that the proposed algorithm possesses superior robustness when handling biomass prediction in complex scenarios. Videos S1–S4 present actual tracking results comparing our method (MFFT) with three other tracking methods (https://drive.google.com/file/d/1rq3JMPvyGqToX7-yekm7z_zgyh8k7LVc3/view?usp=sharing). It is clear that our method achieves the best tracking performance. The MFFT counting error is only 4.30 percent. This is much lower than other baselines such as ByteTrack. The significantly lower counting error can be attributed to two factors. First, the diagonal feature provides stable characterization of mango

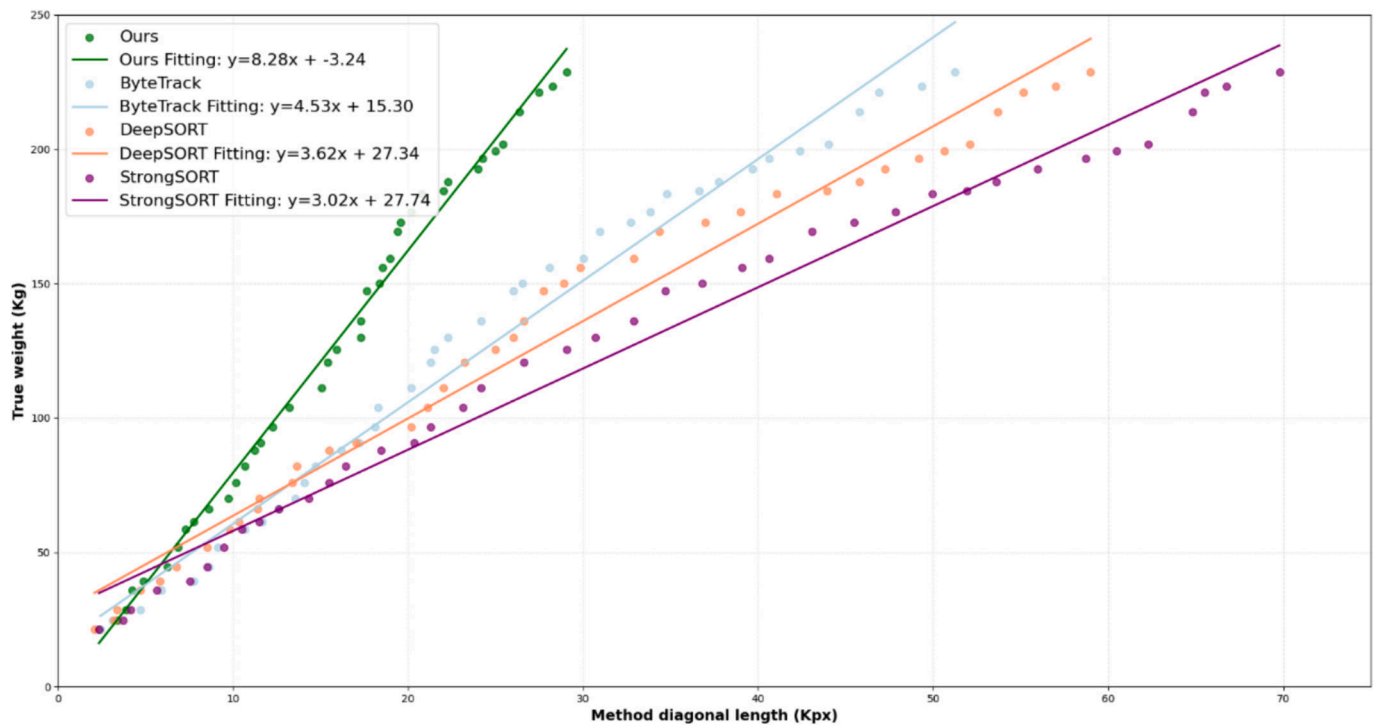
morphology. This effectively alleviates matching deviations caused by viewpoint changes. Second, the ID lifecycle management mechanism filters out miscounts caused by transient missed detections through caching and smoothing strategies. These two mechanisms jointly improve the accuracy of cross-frame association.

For the task of classifying and counting mangoes on trees versus those on the ground, the comparative analysis in Table 5 and Fig. 18 reveals discrepancies in scenario adaptability among different algorithms. Based on video data comprising thirty data points, MFFT achieves regression equations of y equals $1.13x$ minus 10.75 with an R

Table 4

Comparison of our method with count/biomass estimates from three other popular tracking algorithms.

Method	Regression equation		FPS	R ²	RMSE	MAE	MRE (%)
Ours (MFFT)	Counting	$y = 1.04x - 4.08$	13.26	0.98	12.95	8.75	4.30
	Biomass estimation	$y = 8.28x - 3.24$		0.99	6.09	4.79	4.49
ByteTrack	Counting	$y = 0.65x + 36.88$	31.90	-	113.49	93.78	29.19
	Biomass estimation	$y = 4.53x + 15.30$		0.97	8.99	7.74	8.35
DeepSORT	Counting	$y = 0.55x + 50.65$	20.02	-	165.51	134.95	41.31
	Biomass estimation	$y = 3.62x + 27.34$		0.96	9.88	8.42	10.39
StrongSORT	Counting	$y = 0.50x + 28.20$	16.93	-	248.57	219.70	74.39
	Biomass estimation	$y = 3.02x + 27.74$		0.96	8.47	7.31	10.86

**Fig. 19.** Comparison of biomass estimates from our method versus three other popular tracking algorithms.**Table 5**

Analysis of Results Evaluating Mango Quantities on Trees and on the Ground Using ByteTrack and MFFT Algorithms

Method	Category	Regression equation	R ²	RMSE	MAE	MRE (%)
Ours (MFFT)	On the tree	$y = 1.13x - 10.75$	0.98	8.92	7.29	6.37
	On the ground	$y = 1.01x + 1.01$	0.99	1.10	0.94	3.36
ByteTrack	On the tree	$y = 0.66x - 1.15$	0.92	13.27	10.81	10.78
	On the ground	$y = 0.47x + 11.14$	0.94	4.89	4.07	24.36

squared value of 0.98 for tree mangoes, and y equals $1.01x$ plus 1.01 with an R squared value of 0.99 for ground mangoes. The slopes of both equations are close to 1, and error metrics, such as an MRE of only 3.36 percent for ground mangoes, indicate that the algorithm maintains high precision across both scenarios. In contrast, the ByteTrack algorithm

yields a regression slope of only 0.47 and an MRE as high as 24.36 percent for ground mangoes, reflecting poor adaptability to ground scenarios characterized by occlusion and complex backgrounds. The scatter plot comparison in Fig. 18 demonstrates that MFFT adheres closely to the y equals x baseline in both scenarios, whereas ByteTrack results for ground mangoes deviate significantly from the diagonal, further validating the generalization capability of MFFT in complex environments.

In summary, the series of comparative experiments comprehensively validate the superiority of the proposed MFFT algorithm across multiple dimensions, including global counting accuracy, biomass estimation precision, and adaptability to specific scenarios. MFFT not only

Table 6

Results of MOT Methods in test videos.

	MOTA	MOTP	IDF1	Inference Time (ms)
MFFT(Ours)	0.6513	0.6366	0.7075	75.36
ByteTrack	0.6117	0.6308	0.6684	29.82
DeepSORT	0.5501	0.6260	0.6373	47.02
StrongSORT	0.4995	0.5963	0.5225	58.91

significantly outperforms mainstream comparative algorithms across all core quantitative metrics but also provides a reliable and efficient solution for precise fruit monitoring and yield assessment in complex orchard environments through stable ID management and accurate feature fusion mechanisms.

To compare tracker performance, we evaluated multiple MOT methods on our captured video dataset. We selected ByteTrack, DeepSORT, and StrongSORT as alternative trackers, integrating them with our segmentation network to perform tracking and counting tasks. The experimental results are presented in Table 6. Our method demonstrates superior adaptability and efficacy in tracking and counting mangoes within the complex and dynamic environment of orchards, achieving an MOTA of 0.6513, an MOTP of 0.6366, and an IDF1 of 0.7075. StrongSORT exhibits suboptimal performance with an MOTA of only 0.4995. While ByteTrack demonstrates inferior overall performance compared to MFFT, its inference time of merely 29.82 milliseconds validates its high computational efficiency. Among all methods, our approach incurs the longest inference time, which represents a primary direction for future optimization.

To facilitate superior visualization of the segmentation, tracking, and counting results generated by MFFT, frames were extracted at intervals of ten from a randomly selected output tracking video. A sequence of eight images is presented, commencing at frame t and concluding at frame $t + 70$. As illustrated in Fig. 20, the comprehensive tracking process of our method is demonstrated, fully showcasing mango classification and segmentation, diagonal length calculation, and tracking-based counting. The visualization confirms that our segmentation

approach achieves high fidelity on target mangoes, providing a robust foundation for tracking and counting within complex and dynamic environments. Furthermore, the tracker maintains stable and continuous tracking of target mangoes while responding effectively to the appearance of new instances and the disappearance of existing ones. However, it is noteworthy that within such complex environments, our method exhibits instances of incomplete segmentation and ID switching. These aspects represent critical areas requiring further enhancement in future research.

To verify the improvement of morphological constraints on Kalman filtering, we kept other components and association strategies unchanged under the unified tracking framework, and only compared the traditional Kalman filter with the improved Kalman filter proposed in this study. Tracking performance was evaluated using MOTA, MOTP, and IDF1, and the average single-frame inference time was recorded to quantify the robustness and efficiency of the two filtering methods. The results are shown in Table S3.

As can be seen from Table S3, under the same front-end segmentation and matching strategy, the traditional Kalman filter achieves lower tracking metrics. It relies only on position and scale information, making it difficult to adapt to morphological fluctuations caused by perspective changes, mango posture variations, and occlusions in orchards. This easily leads to prediction deviation, matching errors, and ID switches, resulting in lower MOTA and IDF1. In contrast, the improved Kalman filter integrates diagonal morphological features, which significantly enhances tracking accuracy and ID stability. It only introduces a slight increase in computational cost without compromising real-time

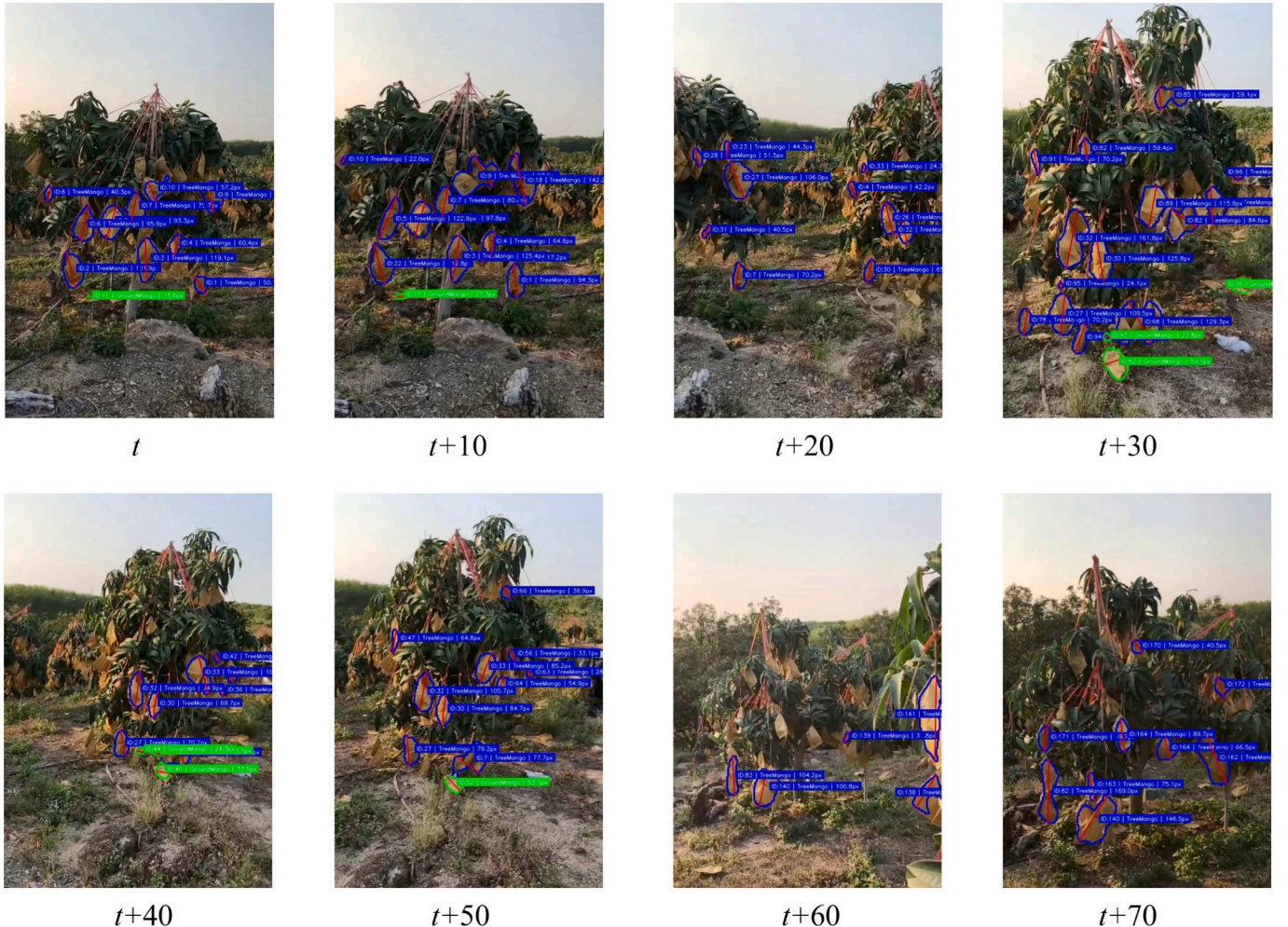


Fig. 20. Visualization results of mango counting in the test video.

Table 7
Compared with other studies.

Object	Network	Performance	Scene complexity	Function	Ref.
Strawberry	Mask R-CNN/ TensorRT	0.83mAP, 25.46 FPS	Sparse planting in the greenhouse	Instance Segmentation	(Crespo et al., 2025)
Apple	MCA-YOLO	0.95 mAP, 0.76 mm error	Moderate shading	Segmentation, 3D positioning	(Zhou, et al., 2025a,2025b)
Mango	YOLO	0.93mAP, 0.76MOTA	Light shading	Testing, tracking	(Pichhika et al., 2025)
Passion fruit	YOLOv8n + OCSORT + CRCM	0.83mAP, 87% accuracy in counting	Moderate shading	Detection, Tracking, Counting	(Tu et al., 2025)
Kiwi fruit	TCA-YOLOv5s	0.87 mAP, 94% accuracy in counting	Light shading	Detection, Tracking, Counting	(Zhang, et al., 2025a,2025b)
Apple	YOLOv8n	0.89 mAP,0.85R2	Moderate shading	Detection, Tracking, Counting	(Zhai et al., 2025)
Mango (tree and ground)	MFFT	mIoU = 0.71; 40.32 FPS; Counting error 4.30%; Biomass estimation error 4.49%;	Extremely shading	Segmentation, Tracking, Counting, Biomass Estimation	This work

performance, thus meeting the requirements of real-time mango tracking in orchards.

3.3. Compared with other studies

Compared with other studies on fruit visual detection in orchards, the MFFT method proposed in this work exhibits superior adaptability to complex scenarios and more comprehensive functional integration (Table 7). Most existing research targets crops like strawberries, apples, passion fruits and kiwifruits, adopting methods such as Mask R-CNN and YOLO to realize single or partial functions including instance segmentation, detection-tracking-counting and segmentation-3D localization, with mAP ranging from 0.82 to 0.95. However, these methods are only suitable for sparse or moderately occluded orchard environments and lack the key function of biomass estimation. For mango detection, the existing YOLO combined with centroid tracking merely achieves simple detection and tracking under mild occlusion, with limited 0.76 MOTA. In contrast, MFFT is specially designed for on-tree and on-ground mangoes in extremely occluded scenarios, integrating segmentation, tracking, counting and biomass estimation into a complete pipeline. It achieves a segmentation mIoU of 0.71 with an inference speed of 40.32 FPS, and controls the global counting relative error at 4.30% and biomass estimation error at only 4.49%, breaking the limitations of existing studies in scenario adaptability and functional completeness.

3.4. Discussion

3.4.1. Advantages

This study presents a comprehensive integrated framework for segmentation, tracking, counting, and biomass estimation tailored to mango orchard environments. Extensive experiments demonstrate that our method yields promising results. The dataset in this study features typical field complexity, including dense mangoes with heavy occlusion and strong interference from bare soil, weeds, branches, leaves, and uneven illumination. All data are collected from real orchards. Accordingly, the proposed method is specially optimized for complex scenes via background gating, boundary attention, and multi-scale feature fusion to suppress noise and reduce false detection. Quantitative results confirm that complex backgrounds significantly affect segmentation and tracking performance, demonstrating the necessity of our dataset and algorithm design.

In the segmentation component, this study employs a custom-designed architecture combining pixel-level semantic segmentation with connected component pseudo-instance extraction. This approach precisely fulfills the core requirements of mango counting and biomass estimation in orchards. The model balances pixel-level and instance-level performance, achieving outstanding metrics such as mIoU and F1 scores in semantic segmentation. By extracting pseudo-instances via connected component analysis, the model achieves a mask mAP approaching that of mainstream instance segmentation models like

Mask R-CNN, providing a sufficient foundation for precise counting tasks.

In the tracking and counting component, the designed tracker exhibits superior stability and accuracy compared to mainstream models like ByteTrack within the complex scenarios of mango orchards. Addressing typical orchard tasks, the tracker distinguishes between on-tree and ground mangoes based on segmentation results, enabling continuous tracking and independent counting for different categories. By fusing multi-dimensional features including position, shape, color, and diagonal morphology, the model maintains stable tracking of individual mango features. Combined with an ID lifecycle management mechanism, this effectively suppresses ID switches and duplicate counting. Furthermore, the tracker supports accuracy analysis based on mango density distribution within single images, allowing for error calibration in high-density clustered regions and low-density isolated regions. This compensates for potential errors in the segmentation module during occlusion, further enhancing overall counting precision to better suit practical operational needs.

In the biomass estimation component, the module is logically highly aligned with core task requirements. Compared to traditional detection box-based counting, this system utilizes diagonal features extracted from segmentation results to construct the estimation model, offering greater practicality and accuracy. Selecting diagonal length over mask area effectively reduces the impact of minor mask fluctuations on biomass estimation and facilitates standard unit calibration. The system explicitly distinguishes between tree-hanging and ground mangoes. Considering that ground mangoes do not directly contribute to yield prediction, separately counting the quantity and biomass of these two categories provides more targeted data support for yield assessment, harvesting scheduling, and management decisions. Additionally, this estimation scheme requires no additional size annotation, integrating seamlessly with segmentation and tracking to realize a unified workflow from segmentation and tracking to biomass estimation, efficiently outputting data required for orchard yield prediction.

Notably, in preliminary experiments, we evaluated mainstream instance segmentation models such as YOLOv8s and DETR but identified inherent limitations that are difficult to circumvent. Visualization of segmentation results on randomly selected images, as shown in Fig. 21, compares our method with DETR and YOLOv8s under identical input conditions, illustrating their respective limitations. Red boxes indicate missed detections, orange boxes indicate over-prediction, and light blue boxes indicate adhesion. Results show that while the YOLOv8 model is efficient and accurate in clear scenarios, it adopts a conservative prediction strategy for mangoes with blurred features, partial occlusion, or uneven lighting, directly discarding such targets and leading to missed detections (Fig. 21b). When paired with a tracker for video processing, YOLOv8 suffers from rough mask shapes and significant latency between mask changes and dynamic video frame variations, compromising tracking coherence. Conversely, DETR exhibits a tendency for over-inference, forcing complete fruit contours based on minimal visible



corners of occluded mangoes (Fig. 21c). This often results in masks being drawn over adjacent mangoes or foliage, causing artificially inflated fruit counts. These defects directly undermine counting accuracy. In contrast, our model effectively circumvents these issues through precise pixel-level segmentation capabilities, providing a reliable basis for counting (Fig. 21a). Compared to the systematic estimation biases caused by YOLOs missed detections or DETRs over-predictions, our model leverages the complete pixel coverage of the segmentation module. Even if diagonal measurements for individual mangoes contain errors due to adhesion, the model can balance these deviations through population characteristics, resulting in estimation results of higher reference value.

3.4.2. Limitations and Potential Improvements

This study acknowledges the following limitations. First, the segmentation component suffers from inherent limitations of semantic segmentation, particularly in high-density scenarios. In conditions of dense mango aggregation and severe mutual occlusion, connected component analysis tends to merge multiple fruits into a single instance. This leads to blurred instance boundaries and an inability to achieve precise individual delineation, thereby compromising the accuracy of diagonal measurements, the reliability of tracking feature matching, and counting precision. As illustrated in Fig. 22, our method is prone to misidentifying clustered mangoes as a single entity and exhibits issues with missed detections and incomplete segmentation under poor lighting conditions. Future improvements should prioritize segmentation accuracy. Potential strategies include adopting a hybrid architecture combining semantic segmentation with a lightweight instance branch, such as integrating CenterNet to assist connected component analysis. Additionally, expanding the dataset to include high-density and extreme lighting scenarios is necessary to enhance generalization. Alternatively, transitioning to a pure instance segmentation model utilizing advanced Transformer architectures with query mechanisms, or improving upon existing methods to mitigate YOLO missed detections or DETR over-predictions, could further reduce errors.

Second, the tracker's performance is heavily reliant on segmentation results, leading to insufficient stability in extreme scenarios. Segmentation errors indirectly cause ID switches and misassignments. Furthermore, relying solely on basic visual features can result in re-identification failures during rapid mango movement or instantaneous full occlusion, leading to decreased tracking precision. Minor latency in module coordination also affects video stream synchronization. To address this, future work should capture richer mango appearance features and integrate advanced ReID methods. Moreover, transforming the separate segmentation and tracking processes into a joint framework with bidirectional feedback mechanisms—where tracking precision informs the segmentation model—could significantly improve performance.

Third, biomass estimation is constrained by the limitations of 2D vision, introducing multiple systematic errors. Relying solely on diagonal length requires establishing a simplified and incomplete correspondence with biomass. Without access to 3D dimensions or spatial poses, measurement biases arise from varying shooting angles and irregular fruit morphologies, resulting in lower accuracy compared to 3D methods. Additionally, diagonal measurement precision is directly affected by mask quality, with edge offsets, flickering, or adhesion causing calculation errors. While the 2D approach could be refined by incorporating more reference features, a more advanced solution involves introducing lightweight binocular camera modules to acquire 3D point cloud data. Combining multi-dimensional 2D features with 3D data would compensate for precision defects, enabling single-tree identification and tree-to-ground ratio statistics to support refined yield estimation.

In summary, the proposed segmentation tracking biomass estimation model holds promising application prospects. All experimental samples in this study were naturally bagged mangoes compliant with orchard

standards. Although bagging alters fruit boundary features, the dataset retains key challenges such as leaf occlusion, fruit overlapping, and uneven illumination, making it highly consistent with real production scenarios and of great engineering application value. We also will add non-bagged data to further improve the generalization ability of the algorithm. Future research should strengthen the integration of its components, conduct experiments across diverse orchard environments, and collect data throughout the entire mango growth cycle to provide precise support for yield prediction and field management. We will also incorporate cross-validation methods and an external validation dataset to improve the model's robustness. In addition, the modular architecture of our method provides a foundation for cross-crop transfer. The attention mechanism in the segmentation module can adapt to the appearance features of different crops. The multi-feature fusion framework in the tracking module can be replaced with crop-specific morphological indicators such as fruit diameter and fruit shape index. We will conduct multi-crop validation experiments in future work to further verify its generalization capability.

4. Conclusion

This study proposes a Multi-Feature Fusion Tracking (MFFT) framework for multi-class mango segmentation, stable ID tracking, and biomass estimation in complex occluded orchard scenarios, addressing the key challenges of low segmentation accuracy, frequent ID switches, and inefficient biomass assessment in existing methods. The MFFT integrates a Swin-T backbone with a custom Transformer decoder and dedicated modules (SFE, SOA, BAM), achieving 0.71 mIoU and 0.73 F1-Score at 40.32 FPS for pixel-level segmentation of on-tree and on-ground mangoes, outperforming mainstream models. For tracking, a multi-feature fusion strategy (IoU, HSV, Hu moments, diagonal length) combined with an enhanced Kalman filter and ID lifecycle management effectively suppresses ID switches, yielding a 4.30% MRE for global counting and superior MOTA (0.6513) and IDF1 (0.7075) compared to ByteTrack and so on. By fusing segmentation-derived diagonal features with tracking results, the framework realizes precise biomass estimation with an MRE of only 4.49% ($R^2 = 0.99$). While MFFT demonstrates robust performance in variable lighting and occlusion, it faces limitations in high-density mango segmentation and 2D-based biomass estimation.

Future work will focus on optimizing the segmentation network for cluster-shaped targets. We will construct a joint segmentation and tracking feedback framework. Binocular vision technology will be integrated to enable three-dimensional biomass assessment. In addition, the modular architecture of our method provides a foundation for cross-crop transfer. The attention mechanism in the segmentation module can adapt to the appearance features of different crops. The multi-feature fusion framework in the tracking module can be replaced with crop-specific morphological indicators such as fruit diameter and fruit shape index. We will conduct multi-crop validation experiments in subsequent studies to further verify its generalization capability.

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the authors used ChatGPT in order to improve the language and readability. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

CRediT authorship contribution statement

Wentao Huang: Writing – original draft, Visualization, Data curation, Conceptualization. **Yingsheng Li:** Writing – original draft, Visualization, Methodology, Conceptualization. **Beike Tan:** Writing – original draft, Visualization. **Marija Brkić Bakarić:** Writing – original draft, Visualization. **Zhiqiang Zhu:** Writing – original draft,

Visualization. **Xiaoshuan Zhang:** Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2026.112149>.

Data availability

Data will be made available on request.

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